Using the PAC2002Tire Model

The PAC2002 Magic-Formula tire model has been developed by MSC.Software according to Tyre and Vehicle Dynamics by Pacejka [1]. PAC2002 is latest version of a Magic-Formula model available in Adams/Tire.

Learn about:

When to Use PAC2002

Magic-Formula (MF) tire models are considered the state-of-the-art for modeling tire-road interaction forces in vehicle dynamics applications. Since 1987, Pacejka and others have published several versions of this type of tire model. The PAC2002 contains the latest developments that have been published in *Tyre and Vehicle Dynamics* by Pacejka [1].

In general, a MF tire model describes the tire behavior for rather smooth roads (road obstacle wavelengths longer than the tire radius) up to frequencies of 8 Hz. This makes the tire model applicable for all generic vehicle handling and stability simulations, including:

- Steady-state cornering
- Single- or double-lane change
- Braking or power-off in a turn
- Split-mu braking tests
- J-turn or other turning maneuvers

- ABS braking, when stopping distance is important (not for tuning ABS control strategies)
- Other common vehicle dynamics maneuvers on rather smooth roads (wavelength of road obstacles must be longer than the tire radius)

For modeling roll-over of a vehicle, you must pay special attention to the overturning moment characteristics of the tire (M_x) and the loaded radius modeling. The last item may not be sufficiently accurate in this model.

The PAC2002 model has proven to be applicable for car, truck, and aircraft tires with camber (inclination) angles to the road not exceeding 15 degrees.

PAC2002 and Previous Magic Formula Models

Compared to previous versions, PAC2002 is backward compatible with all previous versions of PAC2002, MF-Tyre 5.x tire models, and related tire property files.

New Features

The enhancements for PAC2002 in Adams/Tire 2005 r2 are:

- More advanced tire-transient modeling using a contact mass in the contact point with the road. This results in more realistic dynamic tire model response at large slip, low speed, and standstill (usemode > 20).
- Parking torque and turn-slip have been introduced: the torque around the vertical axis due to turning at standstill or at low speed (no need for extra parameters).
- Extended loaded radius modeling (see Contact Point and Normal Load Calculation) are suitable for driving under extreme conditions like roll-over events and racing applications.
- The option to use a nonlinear spline for the vertical tire load-deflection instead of a linear tire stiffness. See Contact Point and Normal Load Calculation.
- Modeling of bottoming of the tire to the road by using another spline for defining the bottoming forces. Learn more about wheel bottoming.
- Online scaling of the tire properties during a simulation; the scaling factors of the PAC2002 can now be changed as a function of time, position, or any other variable in your model dataset. See Online Scaling of Tire Properties.

Modeling of Tire-Road Interaction Forces

For vehicle dynamics applications, an accurate knowledge of tire-road interaction forces is inevitable because the movements of a vehicle primarily depend on the road forces on the tires. These interaction forces depend on both road and tire properties, and the motion of the tire with respect to the road.

In the radial direction, the MF tire models consider the tire to behave as a parallel linear spring and linear damper with one point of contact with the road surface. The contact point is determined by considering the tire and wheel as a rigid disc. In the contact point between the tire and the road, the contact forces in longitudinal and lateral direction strongly depend on the slip between the tire patch elements and the road.

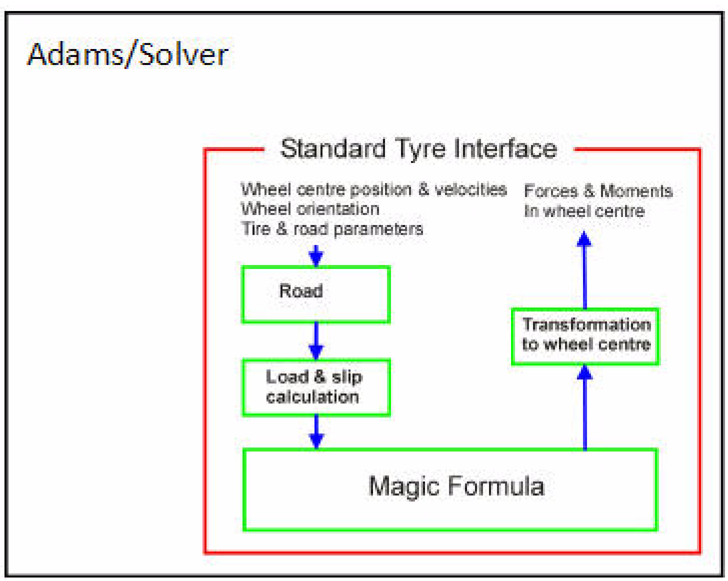
The figure, Input and Output Variables of the Magic Formula Tire Model, presents the input and output vectors of the PAC2002 tire model. The tire model subroutine is linked to the Adams/Solver through the Standard Tire Interface (STI) [3]. The input through the STI consists of:

- Position and velocities of the wheel center
- Orientation of the wheel
- Tire model (MF) parameters
- Road parameters

The tire model routine calculates the vertical load and slip quantities based on the position and speed of the wheel with respect to the road. The input for the Magic Formula consists of the wheel load (F_z), the longitudinal and lateral slip (κ, α), and inclination angle (γ) with the road. The output is the forces (F_x, F_y) and moments (M_x, M_y, M_z) in the contact point between the tire and the road. For calculating these forces, the MF equations use a set of MF parameters, which are derived from tire testing data.

The forces and moments out of the Magic Formula are transferred to the wheel center and returned to Adams/Solver through STI.

Input and Output Variables of the Magic Formula Tire Model



Axis Systems and Slip Definitions

- Axis Systems

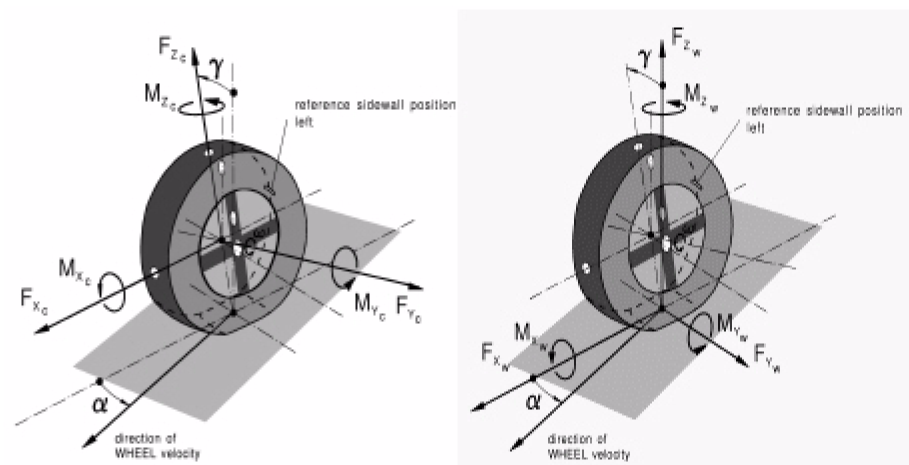
- Units
- Definition of Tire Slip Quantities

Axis Systems

The PAC2002 model is linked to Adams/Solver using the TYDEX STI conventions, as described in the TYDEX-Format [2] and the STI [3].

The STI interface between the PAC2002 model and Adams/Solver mainly passes information to the tire model in the C-axis coordinate system. In the tire model itself, a conversion is made to the W-axis system because all the modeling of the tire behavior as described in this help assumes to deal with the slip quantities, orientation, forces, and moments in the contact point with the TYDEX W-axis system. Both axis systems have the ISO orientation but have different origin as can be seen in *the figure below*.

TYDEX C- and W-Axis Systems Used in PAC2002, Source [2]



The C-axis system is fixed to the wheel carrier with the longitudinal x_c-axis parallel to the road and in the wheel plane (x_c-z_c-plane). The origin of the C-axis system is the wheel center.

The origin of the W-axis system is the road contact-point defined by the intersection of the wheel plane, the plane through the wheel carrier, and the road tangent plane.

The forces and moments calculated by PAC2002 using the MF equations in this guide are in the W-axis system. A transformation is made in the source code to return the forces and moments through the STI to Adams/Solver.

The inclination angle is defined as the angle between the wheel plane and the normal to the road tangent plane (x_w-y_w-plane).

Units

The units of information transferred through the STI between Adams/Solver and PAC2002 are according to the SI unit system. Also, the equations for PAC2002 described in this guide have been developed for use with SI units, although you can easily switch to another unit system in your tire property file. Because of the non-dimensional parameters, only a few parameters have to be changed.

However, the parameters in the tire property file **must always be valid for the TYDEX W-axis system** (ISO oriented). The basic SI units are listed in the table below (also see Definitions).

SI Units Used in PAC2002

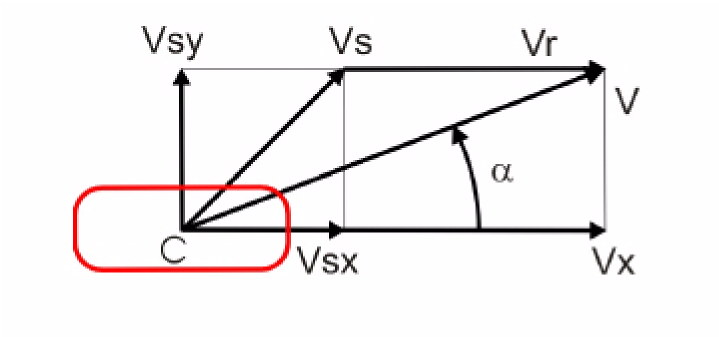
Variable type:	Name:	Abbreviation:	Unit:
Angle	Slip angle Inclination angle	α γ	Radians
Force	Longitudinal force Lateral force Vertical load	F_x F_y F_z	Newton
Moment	Overturning moment Rolling resistance moment Self-aligning moment	M_x M_y M_z	Newton.meter
Speed	Longitudinal speed Lateral speed Longitudinal slip speed Lateral slip speed	V_x V_y V_{sx} V_{sy}	Meters per second
Rotational speed	Tire rolling speed	ω	Radians per second

Definition of Tire Slip Quantities

The longitudinal slip velocity V_{sx} in the contact point (W-axis system, see Slip Quantities at Combined Cornering and Braking/Traction) is defined using the longitudinal speed V_x, the wheel rotational velocity Ω, and the effective rolling radius R_e:

$$V_{sx} = V_x - \Omega R_e \tag{1}$$

Slip Quantities at Combined Cornering and Braking/Traction



The lateral slip velocity is equal to the lateral speed in the contact point with respect to the road plane:

$$V_{sy} = V_y \tag{2}$$

The practical slip quantities κ (longitudinal slip) and α (slip angle) are calculated with these slip velocities in the contact point with:

$$\kappa = -\frac{V_{sx}}{V_x} \tag{3}$$

$$tan\alpha = \frac{V_{sy}}{|V_x|} \tag{4}$$

The rolling speed V_r is determined using the effective rolling radius R_e:

$$V_r = R_e\Omega \tag{5}$$

Turn-slip is one of the two components that form the spin of the tire. Turn-slip ϕ is calculated using the tire yaw velocity $\dot{\psi}$:

$$W_t = \frac{\dot{\psi}}{V_x} \tag{6}$$

The total tire spin φ is calculated using:

$$\psi_1 = \frac{1}{V_x}\{\dot{\psi} - (1 - \varepsilon_\gamma)\Omega sin\gamma\} \tag{7}$$

The total tire spin has contributions of turn-slip and camber. ε_γ denotes the camber reduction factor for the camber to become comparable with turn-slip.

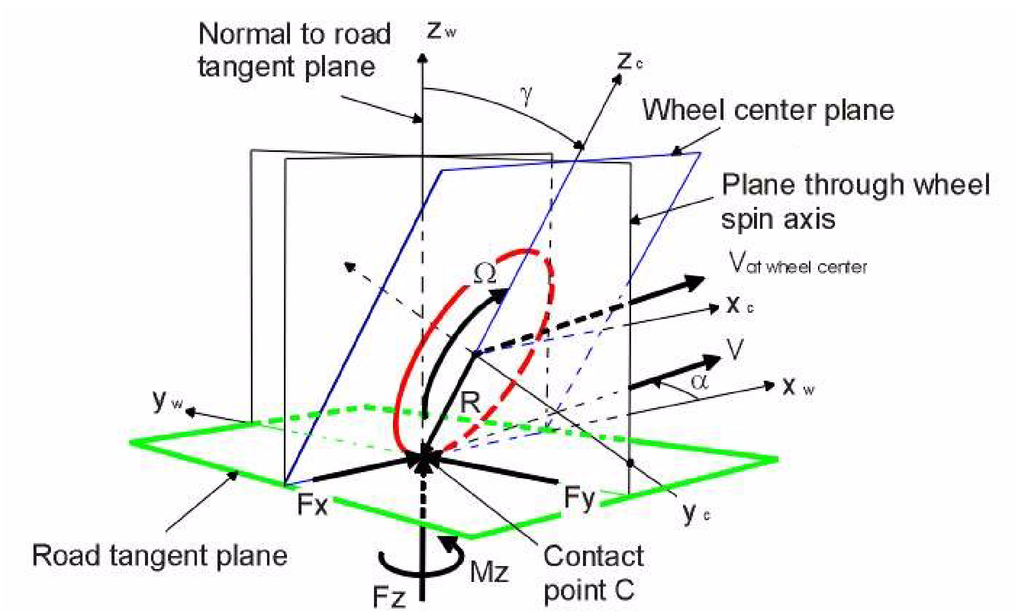
Contact Point and Normal Load Calculation

Contact Point

In the vertical direction, the tire is modeled as a parallel linear spring and damper having one point of contact (C) with the road. This is valid for road obstacles with a wavelength larger than the tire radius (for example, for car tires 1m).

For calculating the kinematics of the tire relative to the road, the road is approximated by its tangent plane at the road point right below the wheel center (see the figure below).

Contact Point C: Intersection between Road Tangent Plane, Spin Axis Plane, and Wheel Plane



The contact point is determined by the line of intersection of the wheel center-plane with the road tangent (ground) plane and the line of intersection of the wheel center-plane with the plane through the wheel spin axis.

The normal load F_z of the tire is calculated with the tire deflection ρ as follows:

$$F_z = \left\{1 + q_{V2}|\Omega|\frac{R_o}{V_o} - \left(q_{Fcxl}\frac{F_x}{F_{z0}}\right)^2 - \left(q_{Fcyl}\frac{F_y}{F_{z0}}\right)^2 + q_{Fc\gamma l}\gamma^2\right\} \tag{8}$$

$$\left[q_{Fzl}\frac{\rho}{R_0} + q_{Fz2}\left(\frac{\rho}{R_0}\right)^2\right]F_{z0} + K_z \cdot \dot{\rho}$$

Using this formula, the vertical tire stiffness increases due to increasing rotational speed Ω and decreases by longitudinal and lateral tire forces. If q_{Fz1} and q_{Fz2} are zero, q_{Fz1} will be defined as $C_z R_0/F_{z0}$.

When you do not provide the coefficients q_{V2}, q_{Fcx}, q_{Fcy}, q_{Fz1}, q_{Fz2} and $q_{Fc\gamma}$ in the tire property file, the normal load calculation is compatible with previous versions of PAC2002, because, in that case, the normal load is calculated using the linear vertical tire stiffness C_z and tire damping K_z according to:

$$F_z = C_z\rho\lambda_{Cz} + K_z\dot{\rho}$$

Instead of the linear vertical tire stiffness C_z (= $q_{Fz1}F_{z0}/R_0$), you can define an arbitrary tire deflection - load curve in the tire property file in the section [DEFLECTION_LOAD_CURVE] (see the Example of PAC2002 Tire Property File). If a section called [DEFLECTION_LOAD_CURVE] exists, the load deflection data points with a cubic spline for inter- and extrapolation are used for the calculation of the vertical force of the tire. Note that you must specify C_z in the tire property file, but it does not play any role.

Loaded and Effective Tire Rolling Radius

With the loaded tire radius Rl defined as the distance of the wheel center to the contact point of the tire with the road, the tire deflection can be calculated using the free tire radius R_0 and a correction for the tire radius growth due to the rotational tire speed Ω:

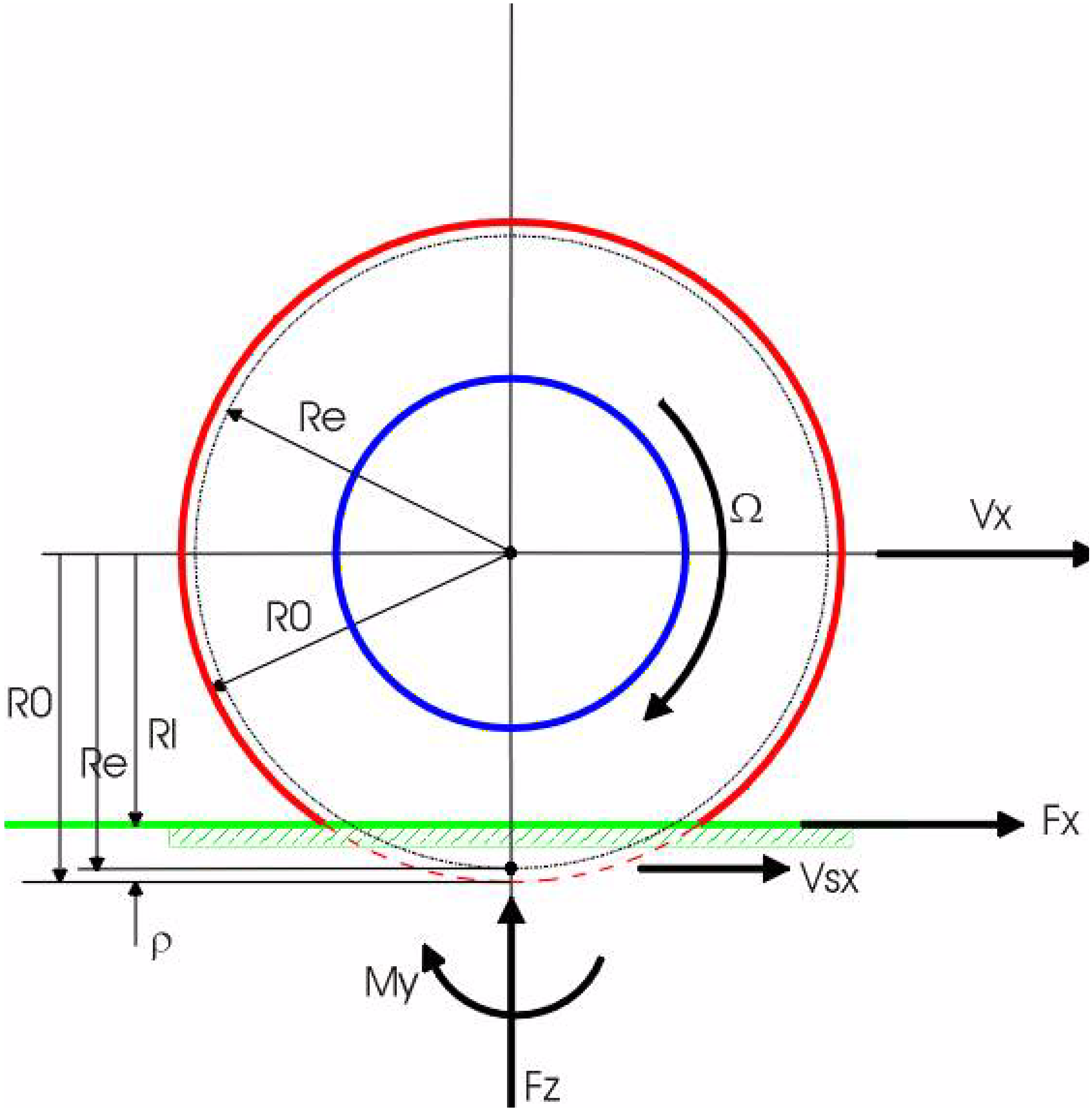
$$\rho = R_0 - R_1 + q_{V1}R_0\left(\Omega\frac{R_0}{V_0}\right)^2 \tag{9}$$

The effective rolling radius R_e (at free rolling of the tire), which is used to calculate the rotational speed of the tire, is defined by:

$$R_e = \frac{V_x}{\Omega} \tag{10}$$

For radial tires, the effective rolling radius is rather independent of load in its load range of operation because of the high stiffness of the tire belt circumference. Only at low loads does the effective tire radius decrease with increasing vertical load due to the tire tread thickness. See the figure below.

Effective Rolling Radius and Longitudinal Slip



To represent the effective rolling radius R_e, a MF-type of equation is used:

$$R_e = R_0 + q_{V1} R_0 \left(\frac{\Omega R_0}{V_0}\right)^2 - \rho_{Fz0}[D_{Reff} arctan(B_{Reff}\rho^d) + F_{Reff}\rho^d] \tag{11}$$

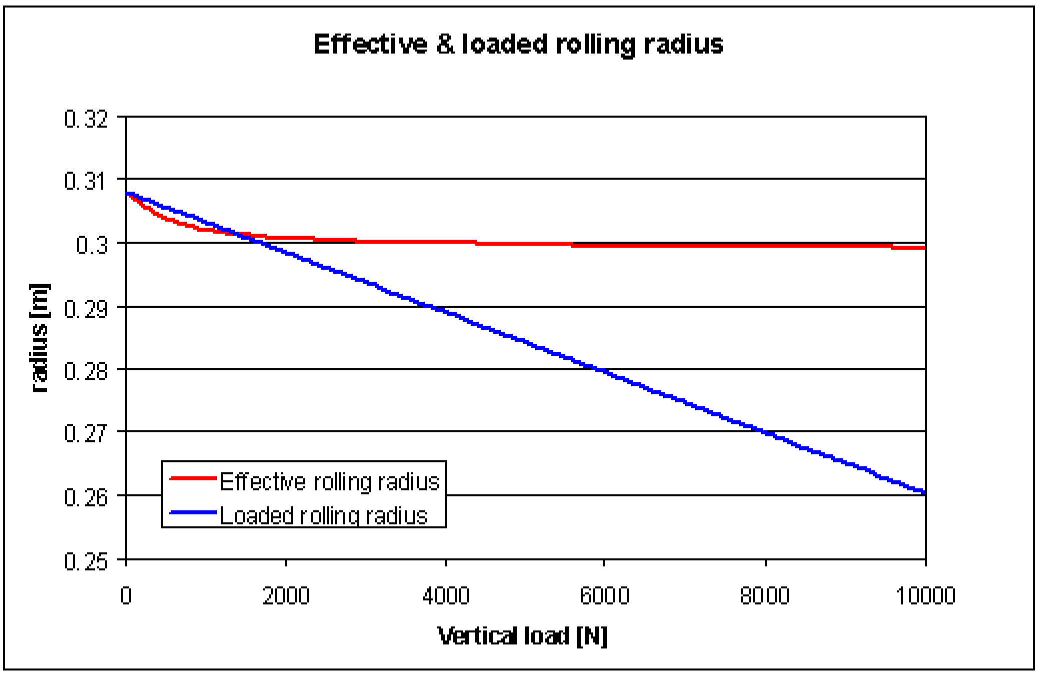
in which ρ_{Fz0} is the nominal tire deflection:

$$\rho_{Fz0} = \frac{F_{z0}}{C_z \lambda_{Cz}} \quad (12)$$

and ρ^d is called the dimensionless radial tire deflection, defined by:

$$\rho^d = \frac{\rho}{\rho_{F_{z0}}} \quad (13)$$

Example of Loaded and Effective Tire Rolling Radius as Function of Vertical Load



Normal Load and Rolling Radius Parameters

Name:	Name Used in Tire Property File:	Explanation:
F_{z0}	FNOMIN	Nominal wheel load
R_o	UNLOADED_RADIUS	Free tire radius
B_{Reff}	BREFF	Low load stiffness effective rolling radius
D_{Reff}	DREFF	Peak value of effective rolling radius
F_{Reff}	FREFF	High load stiffness effective rolling radius
C_z	VERTICAL_STIFFNESS	Tire vertical stiffness (if q_{Fz1}=0)

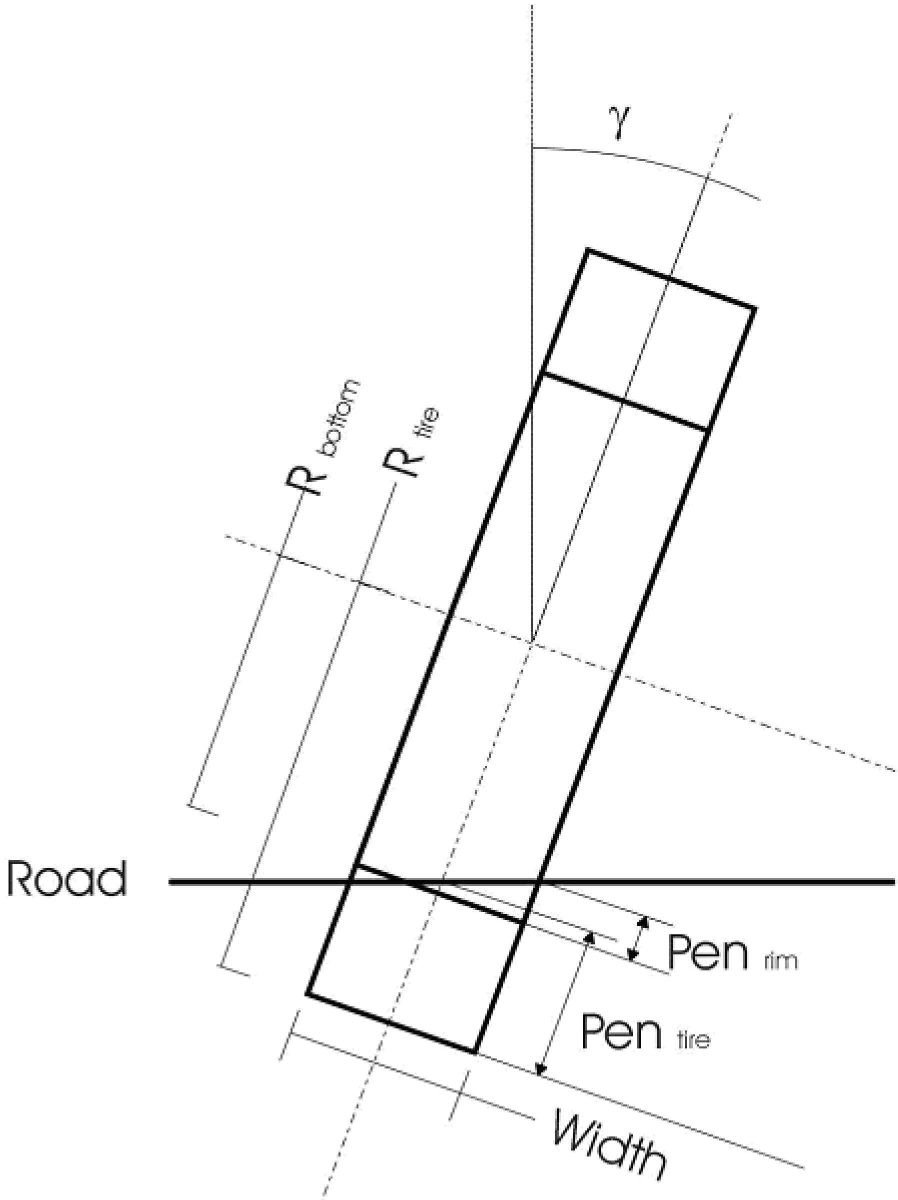
Name:	Name Used in Tire Property File:	Explanation:
K_z	VERTICAL_DAMPING	Tire vertical damping
q_{Fz1}	QFZ1	Tire vertical stiffness coefficient (linear)
q_{Fz2}	QFZ2	Tire vertical stiffness coefficient (quadratic)
q_{Fcx1}	QFCX1	Tire stiffness interaction with Fx
q_{Fcy1}	QFCY1	Tire stiffness interaction with Fy
$q_{Fc\gamma 1}$	QFCG1	Tire stiffness interaction with camber
q_{V1}	QV1	Tire radius growth coefficient
q_{V2}	QV2	Tire stiffness variation coefficient with speed

Wheel Bottoming

You can optionally supply a wheel-bottoming deflection, that is, a load curve in the tire property file in the [BOTTOMING_CURVE] block. If the deflection of the wheel is so large that the rim will be hit (defined by the BOTTOMING_RADIUS parameter in the [DIMENSION] section of the tire property file), the tire vertical load will be increased according to the load curve defined in this section.

Note that the rim-to-road contact algorithm is a simple penetration method (such as the 2D contact) based on the tire-to-road contact calculation, which is strictly valid for only rather smooth road surfaces (the length of obstacles should have a wavelength longer than the tire circumference). The rim-to-road contact algorithm is not based on the 3D-volume penetration method, but can be used in combination with the 3D Contact, which takes into account the volume penetration of the tire itself. If you omit the [BOTTOMING_CURVE] block from a tire property file, no force due to rim road contact is added to the tire vertical force.

You can choose a BOTTOMING_RADIUS larger than the rim radius to account for the tire's material remaining in between the rim and the road, while you can adjust the bottoming load-deflection curve for the change in stiffness.



If (Pen_{tire} - (R_{tire} - R_{bottom}) - ½·width ·| tan(γ) |) < 0, the left or right side of the rim has contact with the road. Then, the rim deflection Pen_{rim} can be calculated using:

$$\delta = \max(0\ ,\ ½ \cdot width \cdot |\tan(\gamma)|) + Pen_{tire} - (R_{tire} - R_{bottom})$$

$$Pen_{rim} = \delta^2 / (2 \cdot width \cdot |\tan(\gamma)|)$$

S_{rim}= ½·width - max(width , δ /| tan(γ) |)/3

with S_{rim}, the lateral offset of the force with respect to the wheel plane.

If the full rim has contact with the road, the rim deflection is:

$Pen_{rim} = Pen_{tire} - (R_{tire} - R_{bottom})$

$S_{rim} = width^2 \cdot |\tan(\gamma)| \cdot /(12 \cdot Pen_{rim})$

Using the load - deflection curve defined in the [BOTTOMING_CURVE] section of the tire property file, the additional vertical force due to the bottoming is calculated, while S_{rim} multiplied by the sign of the inclination γ is used to calculate the contribution of the bottoming force to the overturning moment. Further, the increase of the total wheel load Fz due to the bottoming (F_{zrim}) will not be taken into account in the calculation for F_x, F_y, M_y, and M_z. F_{zrim} will only contribute to the overturning moment M_x using the $F_{zrim} \cdot S_{rim}$.

Note: R_{tire} is equal to the unloaded tire radius R_0; Pen_{tire} is similar to effpen (=ρ).

Basics of the Magic Formula in PAC2002

The Magic Formula is a mathematical formula that is capable of describing the basic tire characteristics for the interaction forces between the tire and the road under several steady-state operating conditions. We distinguish:

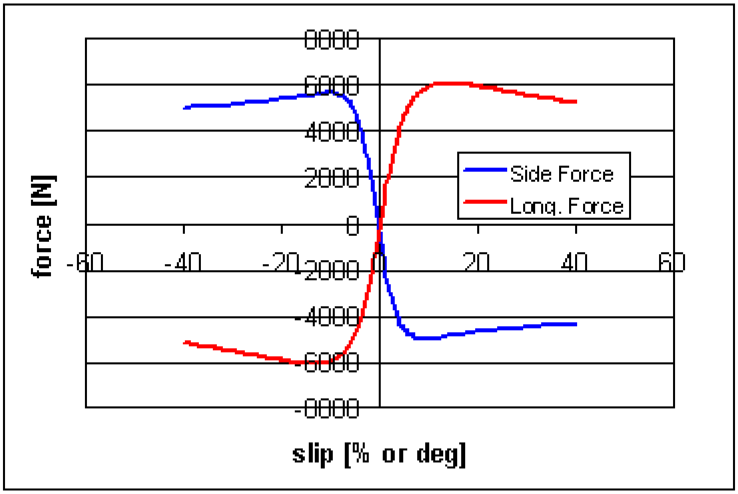
- Pure cornering slip conditions: cornering with a free rolling tire
- Pure longitudinal slip conditions: braking or driving the tire without cornering
- Combined slip conditions: cornering and longitudinal slip simultaneously

For pure slip conditions, the lateral force F_y as a function of the lateral slip α , respectively, and the longitudinal force F_x as a function of longitudinal slip κ , have a similar shape (see the figure, Characteristic Curves for F_x and F_y Under Pure Slip Conditions). Because of the sine - arctangent combination, the basic Magic Formula equation is capable of describing this shape:

$$Y(x) = D\cos[C\arctan\{Bx - E(Bx - \arctan(Bx))\}] \quad (14)$$

where Y(x) is either F_x with x the longitudinal slip κ , or F_y and x the lateral slip α .

Characteristic Curves for F_x and F_y Under Pure Slip Conditions



The self-aligning moment M_z is calculated as a product of the lateral force F_y and the pneumatic trail t added with the residual moment M_{zr}. In fact, the aligning moment is due to the offset of lateral force F_y, called pneumatic trail t, from the contact point. Because the pneumatic trail t as a function of the lateral slip α has a cosine shape, a cosine version the Magic Formula is used:

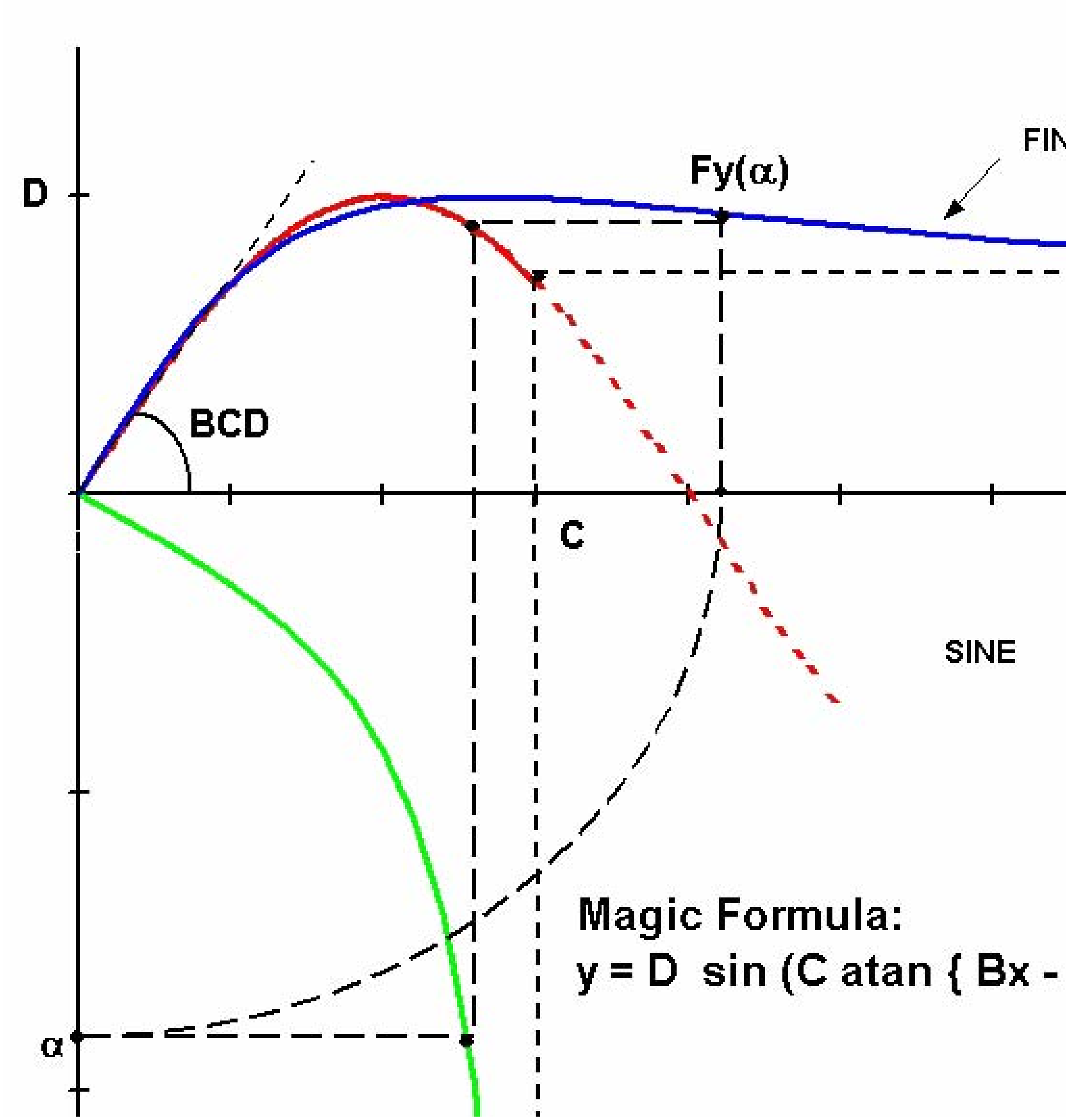
$$Y(x) = D\cos[C\arctan\{Bx - E(Bx - \arctan(Bx))\}] \tag{15}$$

in which Y(x) is the pneumatic trail t as function of slip angle α.

The figure, The Magic Formula and the Meaning of Its Parameters, illustrates the functionality of the B, C, D, and E factor in the Magic Formula:

- D-factor determines the peak of the characteristic, and is called the peak factor.
- C-factor determines the part used of the sine and, therefore, mainly influences the shape of the curve (shape factor).
- B-factor stretches the curve and is called the stiffness factor.
- E-factor can modify the characteristic around the peak of the curve (curvature factor).

The Magic Formula and the Meaning of Its Parameters



In combined slip conditions, the lateral force F_y will decrease due to longitudinal slip or the opposite, the longitudinal force F_x will decrease due to lateral slip. The forces and moments in combined slip conditions are based on the pure slip characteristics multiplied by the so-called weighing functions. Again, these weighting functions have a cosine-shaped MF equation.

The Magic Formula itself only describes steady-state tire behavior. For transient tire behavior (up to 8 Hz), the MF output is used in a stretched string model that considers tire belt deflections instead of slip velocities to cope with standstill situations (zero speed).

Input Variables

The input variables to the Magic Formula are:

Longitudinal slip	κ	[-]
Slip angle	α	[rad]
Inclination angle	γ	[rad]
Normal wheel load	F_z	[N]

Output Variables

Longitudinal force	F_x	[N]
Lateral force	F_y	[N]
Overturning couple	M_x	[Nm]
Rolling resistance moment	M_y	[Nm]
Aligning moment	M_z	[Nm]

The output variables are defined in the W-axis system of TYDEX.

Basic Tire Parameters

All tire model parameters of the model are without dimension. The reference parameters for the model are:

Nominal (rated) load	F_{z0}	[N]
Unloaded tire radius	R_0	[m]
Tire belt mass	m_{belt}	[kg]

As a measure for the vertical load, the normalized vertical load increment df_z is used:

$$df_z = \frac{F_z - F'_{zo}}{F'_{zo}} \tag{16}$$

with the possibly adapted nominal load (using the user-scaling factor, $\lambda_{F_{z0}}$):

$$F'_{zo} = F_{zo} \cdot \lambda_{F_{z0}}$$

Nomenclature of the Tire Model Parameters

In the subsequent sections, formulas are given with non-dimensional parameters a_{ijk} with the following logic:

Tire Model Parameters

Parameter:		Definition:
a =	p	Force at pure slip
	q	Moment at pure slip
	r	Force at combined slip
	s	Moment at combined slip
i =	B	Stiffness factor
	C	Shape factor
	D	Peak value
	E	Curvature factor
	K	Slip stiffness = BCD
	H	Horizontal shift
	V	Vertical shift
	s	Moment at combined slip
	t	Transient tire behavior
j =	x	Along the longitudinal axis
	y	Along the lateral axis
	z	About the vertical axis
k =	1, 2, ...	

User Scaling Factors

A set of scaling factors is available to easily examine the influence of changing tire properties without the need to change one of the real Magic Formula coefficients. The default value of these factors is 1. You can change the factors in the tire property file. The peak friction scaling factors, $\lambda_{\mu\xi}$ and $\lambda_{\mu\psi}$, are also used for the position-dependent friction in 3D Road Contact and 3D Road. An overview of all scaling factors is shown in the following tables.

Scaling Factor Coefficients for Pure Slip

Name:	Name used in tire property file:	Explanation:
λ_{Fzo}	LFZO	Scale factor of nominal (rated) load
λ_{Cz}	LCZ	Scale factor of vertical tire stiffness
λ_{Cx}	LCX	Scale factor of Fx shape factor
$\lambda_{\mu x}$	LMUX	Scale factor of Fx peak friction coefficient
λ_{Ex}	LEX	Scale factor of Fx curvature factor
λ_{Kx}	LKX	Scale factor of Fx slip stiffness
λ_{Hx}	LHX	Scale factor of Fx horizontal shift
λ_{Vx}	LVX	Scale factor of Fx vertical shift
$\lambda_{\gamma x}$	LGAX	Scale factor of inclination for Fx
λ_{Cy}	LCY	Scale factor of Fy shape factor
$\lambda_{\mu y}$	LMUY	Scale factor of Fy peak friction coefficient
λ_{Ey}	LEY	Scale factor of Fy curvature factor
λ_{Ky}	LKY	Scale factor of Fy cornering stiffness
λ_{Hy}	LHY	Scale factor of Fy horizontal shift
λ_{Vy}	LVY	Scale factor of Fy vertical shift
λ_{gy}	LGAY	Scale factor of inclination for Fy
λ_{t}	LTR	Scale factor of peak of pneumatic trail
λ_{Mr}	LRES	Scale factor for offset of residual moment
$\lambda_{\gamma z}$	LGAZ	Scale factor of inclination for Mz
λ_{Mx}	LMX	Scale factor of overturning couple
λ_{VMx}	LVMX	Scale factor of Mx vertical shift
λ_{My}	LMY	Scale factor of rolling resistance moment

Scaling Factor Coefficients for Combined Slip

Name:	Name used in tire property file:	Explanation:
$\lambda_{x\alpha}$	LXAL	Scale factor of alpha influence on Fx
$\lambda_{y\kappa}$	LYKA	Scale factor of alpha influence on Fx
$\lambda_{Vy\kappa}$	LVYKA	Scale factor of kappa-induced Fy
λ_s	LS	Scale factor of moment arm of Fx

Scaling Factor Coefficients for Transient Response

Name:	Name used in tire property file:	Explanation:
$\lambda_{\sigma\kappa}$	LSGKP	Scale factor of relaxation length of Fx
$\lambda_{\sigma\alpha}$	LSGAL	Scale factor of relaxation length of Fy
λ_{gyr}	LGYR	Scale factor of gyroscopic moment

Note that the scaling factors change during the simulation according to any user-introduced function. See the next section, Online Scaling of Tire Properties.

Online Scaling of Tire Properties

PAC2002 can provide online scaling of tire properties. For each scaling factor, a variable should be introduced in the Adams .adm dataset. For example:

```
!lfz0 scaling
!               adams_view_name='TR_Front_Tires until wheel_lfz0_var'
VARIABLE/53
, IC = 1
, FUNCTION = 1.0
```

This lets you change the scaling factor during a simulation as a function of time or any other variable in your model. Therefore, tire properties can change because of inflation pressure, road friction, road temperature, and so on.

You can also use the scaling factors in co-simulations in MATLAB/Simulink.

For more detailed information, see Knowledge Base Article 12732.

Steady-State: Magic Formula in PAC2002

- Steady-State Pure Slip

- Steady-State Combined Slip

Steady-State Pure Slip

- Longitudinal Force at Pure Slip
- Lateral Force at Pure Slip
- Aligning Moment at Pure Slip
- Turn-slip and Parking

Formulas for the Longitudinal Force at Pure Slip

For the tire rolling on a straight line with no slip angle, the formulas are:

$$F_x = F_{x0}(\kappa, F_z, \gamma) \tag{17}$$

$$F_{x0} = D_x sin([C_x arctan\{B_x\kappa_x - E_x(B_x\kappa_x - arctan(B_x\kappa_x))\}] + S_{Vx}) \tag{18}$$

$$\kappa_x = \kappa + S_{Hx} \tag{19}$$

$$\gamma_x = \gamma \cdot \lambda_{\gamma x} \tag{20}$$

with following coefficients:

$$C_x = p_{Cx1} \cdot \lambda_{Cx} \tag{21}$$

$$D_x = \mu_x \cdot F_z \cdot \zeta_1 \tag{22}$$

$$\mu_x = (p_{Dx1} + p_{Dx2}df_z) \cdot (1 - p_{Dx3} \cdot \gamma^2)\lambda_{\mu x} \tag{23}$$

$$E_x = (p_{Ex1} + p_{Ex2}df_z + p_{Ex3}df_z^2) \cdot \{1 - p_{Ex4}sgn(\kappa_x)\} \cdot \lambda_{Ex} \;\; with \;\; E_x \leq 1 \tag{24}$$

the longitudinal slip stiffness:

$$K_x = F_z \cdot (p_{Kx1} + p_{Kx2}df_z) \cdot exp(p_{Kx3}df_z) \cdot \lambda_{Kx} \tag{25}$$

$$K_x = B_x C_x D_x = \frac{\partial F_{x0}}{\partial \kappa_x} at \;\; \kappa_x = 0$$

$$B_x = K_x / (C_x D_x) \tag{26}$$

$$S_{Hx} = (p_{Hx1} + p_{Hx2} \cdot df_z)\lambda_{Hx} \tag{27}$$

$$S_{Vx} = F_z \cdot (p_{Vx1} + p_{Vx2} \cdot df_z)\lambda_{Vx} \cdot \lambda_{\mu x} \cdot \zeta_1 \tag{28}$$

Longitudinal Force Coefficients at Pure Slip

Name:	Name used in tire property file:	Explanation:
p_{Cx1}	PCX1	Shape factor Cfx for longitudinal force
p_{Dx1}	PDX1	Longitudinal friction Mux at Fznom
p_{Dx2}	PDX2	Variation of friction Mux with load
p_{Dx3}	PDX3	Variation of friction Mux with inclination
p_{Ex1}	PEX1	Longitudinal curvature Efx at Fznom
p_{Ex2}	PEX2	Variation of curvature Efx with load
p_{Ex3}	PEX3	Variation of curvature Efx with load squared
p_{Ex4}	PEX4	Factor in curvature Efx while driving
p_{Kx1}	PKX1	Longitudinal slip stiffness Kfx/F_z at Fznom
p_{Kx2}	PKX2	Variation of slip stiffness Kfx/F_z with load
p_{Kx3}	PKX3	Exponent in slip stiffness Kfx/F_z with load
p_{Hx1}	PHX1	Horizontal shift Shx at Fznom
p_{Hx2}	PHX2	Variation of shift Shx with load
p_{Vx1}	PVX1	Vertical shift Svx/F_z at Fznom
p_{Vx2}	PVX2	Variation of shift Svx/F_z with load

Formulas for the Lateral Force at Pure Slip

$$F_y = F_{y0}(\alpha, \gamma, F_z) \tag{29}$$

$$F_{y0} = D_y sin[C_y arctan\{B_y\alpha_y - E_y(B_y\alpha_y - arctan(B_y\alpha_y))\}] + S_{Vy} \tag{30}$$

$$\alpha_y = \alpha + S_{Hy} \tag{31}$$

The scaled inclination angle:

$$\gamma_y = \gamma \cdot \lambda_{\gamma y} \tag{32}$$

with coefficients:

$$C_y = p_{Cy1} \cdot \lambda_{Cy} \tag{33}$$

$$D_y = \mu_y \cdot F_z \cdot \zeta_2 \tag{34}$$

$$\mu_y = (p_{Dy1} + p_{Dy2}df_z) \cdot (1 - p_{Dy3}\gamma_y^2) \cdot \lambda_{\mu y} \tag{35}$$

$$E_y = (p_{Ey1} + p_{Ey2}df_z) \cdot \{1 - (p_{Ey3} + p_{Ey4}\gamma_y)sgn(\alpha_y)\} \cdot \gamma_{Ey} \;\; with \;\; E_y \leq 1 \qquad (36)$$

The cornering stiffness:

$$K_{y0} = P_{Ky1} \cdot F_{z0} \cdot sin\left[2arctan\left\{\frac{F_z}{P_{Ky2}F_0\lambda_{F_{z0}}}\right\}\right] \cdot \lambda_{F_{z0}} \cdot \lambda_{Ky} \qquad (37)$$

$$K_y = K_{y0} \cdot (1 - p_{Ky3}|\gamma_y|) \cdot \zeta_3 \qquad (38)$$

$$B_y = K_y / (C_y D_y) \qquad (39)$$

$$S_{Hy} = (p_{Hy1} + p_{Hy2}df_z) \cdot \lambda_{Hy} + p_{Hy3}\gamma_y \cdot \zeta_0 + \zeta_4 - 1 \qquad (40)$$

$$S_{Vy} = F_z \cdot \{(p_{Vy1} + p_{Vy2}df_z) \cdot \lambda_{Vy} + (p_{Vy3} + p_{Vy4}df_z) \cdot \gamma_y\} \cdot \lambda_{\mu y} \cdot \zeta_4 \qquad (41)$$

The camber stiffness is given by:

$$K_{y\gamma 0} = P_{Hy3}K_{y0} + F_z(p_{Vy3} + p_{Vy4}df_z) \qquad (42)$$

Lateral Force Coefficients at Pure Slip

Name:	Name used in tire property file:	Explanation:
p_{Cy1}	PCY1	Shape factor Cfy for lateral forces
p_{Dy1}	PDY1	Lateral friction Muy
p_{Dy2}	PDY2	Variation of friction Muy with load
p_{Dy3}	PDY3	Variation of friction Muy with squared inclination
p_{Ey1}	PEY1	Lateral curvature Efy at Fznom
p_{Ey2}	PEY2	Variation of curvature Efy with load
p_{Ey3}	PEY3	Inclination dependency of curvature Efy
p_{Ey4}	PEY4	Variation of curvature Efy with inclination
p_{Ky1}	PKY1	Maximum value of stiffness Kfy/Fznom
p_{Ky2}	PKY2	Load at which Kfy reaches maximum value
p_{Ky3}	PKY3	Variation of Kfy/Fznom with inclination
p_{Hy1}	PHY1	Horizontal shift Shy at Fznom
p_{Hy2}	PHY2	Variation of shift Shy with load
p_{Hy3}	PHY3	Variation of shift Shy with inclination
p_{Vy1}	PVY1	Vertical shift in Svy/Fz at Fznom

Name:	Name used in tire property file:	Explanation:
p_{Vy2}	PVY2	Variation of shift Svy/Fz with load
p_{Vy3}	PVY3	Variation of shift Svy/Fz with inclination
p_{Vy4}	PVY4	Variation of shift Svy/Fz with inclination and load

Formulas for the Aligning Moment at Pure Slip

$$M'_z = M_{z0}(\alpha, \gamma, F_z) \tag{43}$$

$$M_{z0} = -t \cdot F_{y0} + M_{zr}$$

with the pneumatic trail t:

$$t(\alpha_t) = D_t cos[C_t arctan\{B_t\alpha_t - E_t(B_t\alpha_t - arctan(B_t\alpha_t))\}]cos(\alpha) \tag{44}$$

$$\alpha_t = \alpha + S_{Ht} \tag{45}$$

and the residual moment M_{zr}:

$$M_{zr}(\alpha_r) = D_r cos[C_r arctan(B_r\alpha_r)] \cdot cos(\alpha) \tag{46}$$

$$\alpha_r = \alpha + S_{Hf} \tag{47}$$

$$S_{Hf} = S_{Hy} + S_{Vy}/K_y \tag{48}$$

The scaled inclination angle:

$$\gamma_z = \gamma \cdot \lambda_{\gamma z} \tag{49}$$

with coefficients:

$$B_t = (q_{Bz1} + q_{Bz2}df_z + q_{Bz3}df_z^2) \cdot (1 + q_{Bz4}\gamma_z + q_{Bz5}|\gamma_z|) \cdot \lambda_{Ky}/\lambda_{\mu y} \tag{50}$$

$$C_t = q_{Cz1} \tag{51}$$

$$D_t = F_z \cdot (q_{Dz1} + q_{Dz2}df_z) \cdot (1 + q_{Dz3}\gamma_z + q_{Dz4}\gamma_z^2) \cdot \frac{R_0}{F_{z0}} \cdot \lambda_t \cdot \zeta_5 \tag{52}$$

$$E_t = (q_{Ez1} + q_{Ez2}df_z + q_{Ez3}df_z^2) \tag{53}$$

$$\left\{1 + (q_{Ez4} + q_{Ez5}\gamma_z)\left(\left(\frac{2}{\pi}\right) \cdot arctan(B_t \cdot C_t \cdot \alpha_t)\right)\right\} \; with \; (E_t \leq 1)$$

$$S_{Ht} = q_{Hz1} + q_{Hz2}df_z + (q_{Hz3} + q_{Hz4} \cdot df_z)\gamma_z \tag{54}$$

$$B_r = \left(q_{Bz9} \cdot \frac{\lambda_{Ky}}{\lambda_{\mu y}} + q_{Bz10} \cdot B_y \cdot C_y\right) \cdot \zeta_6 \tag{55}$$

$$C_r = \zeta_7$$

$$D_r = F_z \cdot [(q_{Dz6} + q_{Dz7}df_z) \cdot \lambda_r + (q_{Dz8} + q_{Dz9}df_z) \cdot \gamma_z] \cdot R_o \cdot \lambda_{\mu\gamma} + \zeta_8 - 1 \tag{56}$$

An approximation for the aligning moment stiffness reads:

$$K_z = -t \cdot K_y \qquad \left(\approx -\frac{\partial M_z}{\partial \alpha} \; at \; \alpha\right) = 0) \tag{57}$$

Aligning Moment Coefficients at Pure Slip

Name:	Name used in tire property file:	Explanation:
q_{Bz1}	QBZ1	Trail slope factor for trail Bpt at Fznom
q_{Bz2}	QBZ2	Variation of slope Bpt with load
q_{Bz3}	QBZ3	Variation of slope Bpt with load squared
q_{Bz4}	QBZ4	Variation of slope Bpt with inclination
q_{Bz5}	QBZ5	Variation of slope Bpt with absolute inclination
q_{Bz9}	QBZ9	Slope factor Br of residual moment Mzr
q_{Bz10}	QBZ10	Slope factor Br of residual moment Mzr
q_{Cz1}	QCZ1	Shape factor Cpt for pneumatic trail
q_{Dz1}	QDZ1	Peak trail Dpt = Dpt*(Fz/Fznom*R0)
q_{Dz2}	QDZ2	Variation of peak Dpt with load
q_{Dz3}	QDZ3	Variation of peak Dpt with inclination
q_{Dz4}	QDZ4	Variation of peak Dpt with inclination squared.
q_{Dz6}	QDZ6	Peak residual moment Dmr = Dmr/ (Fz*R0)
q_{Dz7}	QDZ7	Variation of peak factor Dmr with load
q_{Dz8}	QDZ8	Variation of peak factor Dmr with inclination

Name:	Name used in tire property file:	Explanation:
q_{Dz9}	QDZ9	Variation of Dmr with inclination and load
q_{Ez1}	QEZ1	Trail curvature Ept at Fznom
q_{Ez2}	QEZ2	Variation of curvature Ept with load
q_{Ez3}	QEZ3	Variation of curvature Ept with load squared
q_{Ez4}	QEZ4	Variation of curvature Ept with sign of Alpha-t
q_{Ez5}	QEZ5	Variation of Ept with inclination and sign Alpha-t
q_{Hz1}	QHZ1	Trail horizontal shift Sht at Fznom
q_{Hz2}	QHZ2	Variation of shift Sht with load
q_{Hz3}	QHZ3	Variation of shift Sht with inclination
q_{Hz4}	QHZ4	Variation of shift Sht with inclination and load

Turn-slip and Parking

For situations where turn-slip may be neglected and camber remains small, the reduction factors ζ_i that appear in the equations for steady-state pure slip, are to be set to 1:

$$\zeta_i = 1 \qquad i = 0. \qquad 1....8$$

For larger values of spin, the reduction factors are given below.

The weighting function ζ_1 is used to let the longitudinal force diminish with increasing spin, according to:

$$\zeta_i = cos[arc\,tan(B_{x\varphi}R_0\varphi)]$$

with:

$$B_{x\varphi} = p_{Dx\varphi 1}(1 + p_{Dx\varphi 2}df_z)cos[arc\,tan(p_{Dx\varphi 3}\kappa)]$$

The peak side force reduction factor ζ_2 reads:

$$\zeta_2 = cos[arc\,tan\{B_{y\varphi}(R_0|\varphi| + p_{Dy\varphi 4}\sqrt{R_0|\varphi|})\}]$$

with:

$$B_{y\varphi} = p_{Dx\varphi 1}(1 + p_{Dx\varphi 2}df_z)cos[arc\,tan(p_{Dx\varphi 3}tan\alpha)]$$

The cornering stiffness reduction factor ζ_3 is given by:

$$\zeta_3 = cos[arctan(p_{Ky\varphi 1}R_0^2\varphi^2)]$$

The horizontal shift of the lateral force due to spin is given by:

$$S_{Hy\varphi} = D_{Hy\varphi}sin[C_{Hy\varphi}arctan\{B_{Hy\varphi}R_o\varphi - E_{Hy\varphi}(B_{Hy\varphi}R_0\varphi - arctan(B_{Hy\varphi}R_0\varphi))\}]$$

The factors are defined by:

$$C_{Hy\varphi} = p_{Hy\varphi 1}$$

$$D_{Hy\varphi} = (p_{Hy\varphi 2} + p_{Hy\varphi 3}df_z) \cdot sin(V_x)$$

$$E_{Hy\varphi} = P_{Hy\varphi 4}$$

$$B_{Hy\varphi} = \frac{K_{yR\varphi 0}}{C_y D_y K_{y0}}$$

The spin force stiffness $K_{yR\varphi 0}$ is related to the camber stiffness $K_{y\gamma 0}$:

$$K_{yR\varphi 0} = \frac{K_{y\gamma 0}}{1 - \varepsilon_\gamma}$$

in which the camber reduction factor is given by:

$$\varepsilon_\gamma = p_{\varepsilon\gamma\varphi 1}(1 + p_{\varepsilon\gamma\varphi 2}df_z)$$

The reduction factors ζ_0 and ζ_4 for the vertical shift of the lateral force are given by:

$$\zeta_0 = 0$$

$$\zeta_4 = 1 + S_{Hy\varphi} - S_{Vy\gamma}/K_y$$

The reduction factor for the residual moment reads:

$$\zeta_8 = 1 + D_{r\varphi}$$

The peak spin torque $Dr\varphi$ is given by:

$$D_{r\varphi} = D_{Dr\varphi}sine[C_{Dr\varphi}arctan\{B_{Dr\varphi}R_0\varphi - E_{Dr\varphi}(B_{Dr\varphi}R_0\varphi - arctan(B_{Dr\varphi}R_0\varphi))\}]$$

The maximum value is given by:

$$D_{Dr\varphi} = \frac{M_{z\varphi\infty}}{sin\left(\frac{\pi}{2}C_{Dr\varphi}\right)}$$

The pneumatic trail reduction factor due to turn slip is given by:

$$\xi_5 = cos[arc\,tan(q_{Dt\varphi1}R_0\varphi)]$$

The moment at vanishing wheel speed at constant turning is given by:

$$M_{z\varphi\infty} = q_{Cr\varphi1}\mu_y R_0 F_z\sqrt{F_z/F_{z0}}$$

The shape factors are given by:

$$C_{Dr\varphi} = q_{Dr\varphi1}$$

$$E_{Dr\varphi} = q_{Dr\varphi2}$$

$$B_{Dr\varphi} = \frac{K_{z\gamma r0}}{C_{Dr\varphi}D_{Dr\varphi}(1-\varepsilon_y)}$$

in which:

$$K_{z\gamma r0} = F_z R_0(q_{Dz8} + q_{Dz9}df_z)$$

The reduction factor ζ_6 reads:

$$\zeta_6 = cos[arc\,tan(q_{Br\varphi1}R_0\varphi)]$$

The spin moment at 90º slip angle is given by:

$$M_{z\varphi90} = M_{z\varphi\infty}\cdot\frac{2}{\pi}\cdot arc\,tan(q_{Cr\varphi2}R_0|\varphi|)\cdot G_{yx}(\kappa)$$

The spin moment at 90º slip angle is multiplied by the weighing function $G_{y\kappa}$ to account for the action of the longitudinal slip (see steady-state combined slip equations).

The reduction factor ζ_7 is given by:

$$\zeta_7 = \frac{2}{\pi} \cdot arccos[M_{z\varphi 90} / |D_{Dr\varphi}|]$$

Turn-Slip and Parking Parameters

Name:	Name used in tire property file:	Explanation:
$p\varepsilon\gamma\varphi 1$	PECP1	Camber spin reduction factor parameter in camber stiffness
$p\varepsilon\gamma\varphi 2$	PECP2	Camber spin reduction factor varying with load parameter in camber stiffness
$pDx\varphi 1$	PDXP1	Peak Fx reduction due to spin parameter
$pDx\varphi 2$	PDXP2	Peak Fx reduction due to spin with varying load parameter
$pDx\varphi 3$	PDXP3	Peak Fx reduction due to spin with kappa parameter
$pDy\varphi 1$	PDYP1	Peak Fy reduction due to spin parameter
$pDy\varphi 2$	PDYP2	Peak Fy reduction due to spin with varying load parameter
$pDy\varphi 3$	PDYP3	Peak Fy reduction due to spin with alpha parameter
$pDy\varphi 4$	PDYP4	Peak Fy reduction due to square root of spin parameter
$pKy\varphi 1$	PKYP1	Cornering stiffness reduction due to spin
$pHy\varphi 1$	PHYP1	Fy-alpha curve lateral shift limitation
$pHy\varphi 2$	PHYP2	Fy-alpha curve maximum lateral shift parameter
$pHy\varphi 3$	PHYP3	Fy-alpha curve maximum lateral shift varying with load parameter
$pHy\varphi 4$	PHYP4	Fy-alpha curve maximum lateral shift parameter
$qDt\varphi 1$	QDTP1	Pneumatic trail reduction factor due to turn slip parameter
$qBr\varphi 1$	QBRP1	Residual (spin) torque reduction factor parameter due to side slip
$qCr\varphi 1$	QCRP1	Turning moment at constant turning and zero forward speed parameter
$qCr\varphi 2$	QCRP2	Turn slip moment (at alpha=90deg) parameter for increase with spin
$qDr\varphi 1$	QDRP1	Turn slip moment peak magnitude parameter
$qDr\varphi 2$	QDRP2	Turn slip moment peak position parameter

The tire model parameters for turn-slip and parking are estimated automatically. In addition, you can specify each parameter individually in the tire property file (see example).

See KB-article ## for further details about parking by means of an example.

Steady-State Combined Slip

PAC2002 has two methods for calculating the combined slip forces and moments. If the user supplies the coefficients for the combined slip cosine 'weighing' functions, the combined slip is calculated according to Combined slip with cosine 'weighing' functions (standard method). If no coefficients are supplied, the so-called friction ellipse is used to estimate the combined slip forces and moments, see section Combined Slip with friction ellipse

Combined slip with cosine 'weighing' functions

- Longitudinal Force at Combined Slip
- Lateral Force at Combined Slip
- Aligning Moment at Combined Slip
- Overturning Moment at Pure and Combined Slip
- Rolling Resistance Moment at Pure and Combined Slip

Formulas for the Longitudinal Force at Combined Slip

$$F_x = F_{x0} \cdot G_{x\alpha}(\alpha, \kappa, F_z) \tag{58}$$

with $G_{x\alpha}$ the weighting function of the longitudinal force for pure slip.

We write:

$$F_x = D_{x\alpha} cos[C_{x\alpha} arctan\{B_{x\alpha}\alpha_s - E_{x\alpha}(B_{x\alpha}\alpha_s - arctan(B_{x\alpha}\alpha_s))\}] \tag{59}$$

$$\alpha_s = \alpha + S_{Hx\alpha} \tag{60}$$

with coefficients:

$$B_{x\alpha} = r_{Bx1} cos[arctan\{r_{Bx2}\kappa\}] \cdot \lambda_{x\alpha} \tag{61}$$

$$C_{x\alpha} = r_{Cx1} \tag{62}$$

$$D_{x\alpha} = \frac{F_{xo}}{cos[C_{x\alpha} arctan\{B_{x\alpha}S_{Hx\alpha} - E_{x\alpha}(B_{x\alpha}S_{Hx\alpha} - arctan(B_{x\alpha}S_{Hx\alpha}))\}]} \tag{63}$$

$$E_{x\alpha} = r_{Ex1} + r_{Ex2} df_z \ \ with \ \ E_{x\alpha} \le 1 \tag{64}$$

$$S_{Hx\alpha} = r_{Hx1} \tag{65}$$

The weighting function follows as:

$$G_{x\alpha} = \frac{cos[C_{x\alpha} arctan\{B_{x\alpha}\alpha_s - E_{x\alpha}(B_{x\alpha}\alpha_s - arctan(B_{x\alpha}\alpha_s))\}]}{cos[C_{x\alpha} arctan[B_{x\alpha}S_{Hx\alpha} - E_{x\alpha}(B_{x\alpha}S_{Hx\alpha} - arctan(B_{x\alpha}S_{Hx\alpha}))]]} \tag{66}$$

Longitudinal Force Coefficients at Combined Slip

Name:	Name used in tire property file:	Explanation:
r_{Bx1}	RBX1	Slope factor for combined slip Fx reduction
r_{Bx2}	RBX2	Variation of slope Fx reduction with kappa
r_{Cx1}	RCX1	Shape factor for combined slip Fx reduction
r_{Ex1}	REX1	Curvature factor of combined Fx
r_{Ex2}	REX2	Curvature factor of combined Fx with load
r_{Hx1}	RHX1	Shift factor for combined slip Fx reduction

Formulas for Lateral Force at Combined Slip

$$F_y = F_{y0} \cdot G_{y\kappa}(\alpha, \kappa, \gamma, F_z) + S_{Vy\kappa} \tag{67}$$

with G_{yk} the weighting function for the lateral force at pure slip and S_{Vyk} the 'κ-induced' side force; therefore, the lateral force can be written as:

$$F_y = D_{y\kappa} cos[C_{y\kappa} arctan\{B_{y\kappa}\kappa_s - E_{y\kappa}(B_{y\kappa}\kappa_s - arctan(B_{y\kappa}\kappa_s))\}] + S_{Vy\kappa} \tag{68}$$

$$\kappa_s = \kappa + S_{Hy\kappa} \tag{69}$$

with the coefficients:

$$B_{y\kappa} = r_{By1} cos[arctan\{r_{By2}(\alpha - r_{By3})\}] \cdot \lambda_{y\kappa} \tag{70}$$

$$C_{y\kappa} = r_{Cy1} \tag{71}$$

$$D_{y\kappa} = \frac{F_{yo}}{cos[C_{y\kappa} arctan\{B_{y\kappa}S_{Hy\kappa} - E_{y\kappa}(B_{y\kappa}S_{Hy\kappa} - arctan(B_{y\kappa}S_{Hy\kappa}))\}]} \tag{72}$$

$$E_{y\kappa} = r_{Ey1} + r_{Ey2} df_z \;\; with \;\; E_{y\kappa} \le 1 \tag{73}$$

$$S_{Hy\kappa} = r_{Hy1} + r_{Hy2} df_z \tag{74}$$

$$S_{Vy\kappa} = D_{Vy\kappa} sin[r_{Vy5} arctan(r_{Vy6}\kappa)] \cdot \lambda_{Vy\kappa} \tag{75}$$

$$D_{Vy\kappa} = \mu_y F_z \cdot (r_{Vy1} + r_{Vy2} df_z + r_{Vy3}\gamma) \cdot cos[arctan(r_{Vy4}\alpha)] \tag{76}$$

The weighting function appears is defined as:

$$G_{y\kappa} = \frac{\cos[C_{y\kappa} \arctan\{B_{y\kappa}\kappa_s - E_{y\kappa}(B_{y\kappa}\kappa_s - \arctan(B_{y\kappa}\kappa_s))\}]}{\cos[C_{y\kappa} \arctan\{B_{y\kappa}S_{Hy\kappa} - E_{y\kappa}(B_{y\kappa}S_{Hy\kappa} - \arctan(B_{y\kappa}S_{Hy\kappa}))\}]} \quad (77)$$

Lateral Force Coefficients at Combined Slip

Name:	Name used in tire property file:	Explanation:
r_{By1}	RBY1	Slope factor for combined Fy reduction
r_{By2}	RBY2	Variation of slope Fy reduction with alpha
r_{By3}	RBY3	Shift term for alpha in slope Fy reduction
r_{Cy1}	RCY1	Shape factor for combined Fy reduction
r_{Ey1}	REY1	Curvature factor of combined Fy
r_{Ey2}	REY2	Curvature factor of combined Fy with load
r_{Hy1}	RHY1	Shift factor for combined Fy reduction
r_{Hy2}	RHY2	Shift factor for combined Fy reduction with load
r_{Vy1}	RVY1	Kappa induced side force Svyk/Muy*Fz at Fznom
r_{Vy2}	RVY2	Variation of Svyk/Muy*Fz with load
r_{Vy3}	RVY3	Variation of Svyk/Muy*Fz with inclination
r_{Vy4}	RVY4	Variation of Svyk/Muy*Fz with alpha
r_{Vy5}	RVY5	Variation of Svyk/Muy*Fz with kappa
r_{Vy6}	RVY6	Variation of Svyk/Muy*Fz with atan (kappa)

Formulas for Aligning Moment at Combined Slip

$$M' = -t \cdot F'_y + M_{zr} + s \cdot F_x \quad (78)$$

with:

$$t = t(\alpha_{t,eq}) \quad (79)$$

$$= D_t \cos[C_t \arctan\{B_t\alpha_{t,eq} - E_t(B_t\alpha_{t,eq} - \arctan(B_t\alpha_{t,eq}))\}]\cos(\alpha) \quad (80)$$

$$F'_{y,\gamma=0} = F_y - S_{Vy\kappa} \quad (81)$$

$$M_{zr} = M_{zr}(\alpha_{r,eq}) = D_r \cos[\arctan(B_r\alpha_{r,eq})]\cos(\alpha) \quad (82)$$

$$t = t(\alpha_{t,eq}) \tag{83}$$

with the arguments:

$$\alpha_{t,eq} = \arctan\sqrt{\tan^2\alpha_t + \left(\frac{K_x}{K_y}\right)^2 \kappa^2} \cdot sgn(\alpha_t) \tag{84}$$

$$\alpha_{r,eq} = \arctan\sqrt{\tan^2\alpha_r + \left(\frac{K_x}{K_y}\right)^2 \kappa^2} \cdot sgn(\alpha_r) \tag{85}$$

Aligning Moment Coefficients at Combined Slip

Name:	Name used in tire property file:	Explanation:
s_{sz1}	SSZ1	Nominal value of s/R0 effect of Fx on Mz
s_{sz2}	SSZ2	Variation of distance s/R0 with Fy/Fznom
s_{sz3}	SSZ3	Variation of distance s/R0 with inclination
s_{sz4}	SSZ4	Variation of distance s/R0 with load and inclination

Formulas for Overturning Moment at Pure and Combined Slip

For the overturning moment, the formula reads both for pure and combined slip situations:

$$M_x = R_o \cdot F_z \cdot \left\{ q_{Sx1}\lambda_{VMx} - q_{Sx2} \cdot \gamma + q_{Sx3} \cdot \frac{F_y}{F_{z0}} \right\} \lambda_{Mx} \tag{86}$$

Overturning Moment Coefficients

Name:	Name used in tire property file:	Explanation:
q_{sx1}	QSX1	Lateral force induced overturning couple
q_{sx2}	QSX2	Inclination induced overturning couple
q_{sx3}	QSX3	Fy induced overturning couple

Formulas for Rolling Resistance Moment at Pure and Combined Slip

The rolling resistance moment is defined by:

$$M_y = R_o \cdot F_z \cdot \{ q_{Sy1} + q_{Sy3} F_x / F_{z0} + q_{Sy3} |V_x / V_{ref}| + q_{Sy4} (V_x / V_{ref})^4 \} \tag{87}$$

If q_{sy1} and q_{sy2} are both zero and FITTYP is equal to 5 (MF-Tyre 5.0), then the rolling resistance is calculated according to an old equation:

$$M_y = R_0(S_{Vx} + K_x \cdot S_{Hx}) \tag{88}$$

Rolling Resistance Coefficients

Name:	Name used in tire property file:	Explanation:
q_{sy1}	QSY1	Rolling resistance moment coefficient
q_{sy2}	QSY2	Rolling resistance moment depending on Fx
q_{sy3}	QSY3	Rolling resistance moment depending on speed
q_{sy4}	QSY4	Rolling resistance moment depending on speed^4
V_{ref}	LONGVL	Measurement speed

Combined Slip with friction ellipse

In case the tire property file does not contain the coefficients for the 'standard' combined slip method (cosine 'weighing functions), the friction ellipse method is used, as described in this section. Note that the method employed here is not part of one of the Magic Formula publications by Pacejka, but is an in-house development of MSC.Software.

$$\kappa_c = \kappa + S_{Hx} + \frac{S_{Vx}}{K_x}$$

$$\alpha_c = \alpha + S_{Hy} + \frac{S_{Vy}}{K_y}$$

$$\alpha* = sin(\alpha_c)$$

$$\beta = acos\left(\frac{|\kappa_c|}{\sqrt{\kappa_c^2 + \alpha*^2}}\right)$$

The following friction coefficients are defined:

$$\mu_{x,\,act} = \frac{F_{x,\,0} - S_{Vx}}{F_z} \qquad \mu_{y,\,act} = \frac{F_{y,\,0} - S_{Vy}}{F_z}$$

$$\mu_{x,\,max} = \frac{D_x}{F_z} \qquad \mu_{y,\,max} = \frac{D_y}{F_z}$$

$$\mu_x = \frac{1}{\sqrt{\left(\frac{1}{\mu_{x,act}}\right)^2 + \left(\frac{tan\beta}{\mu_{y,max}}\right)^2}}$$

$$\mu_y = \frac{tan\beta}{\sqrt{\left(\frac{1}{\mu_{x,max}}\right)^2 + \left(\frac{tan\beta}{\mu_{y,act}}\right)^2}}$$

The forces corrected for the combined slip conditions are:

$$F_x = \frac{\mu_x}{\mu_{x,act}} F_{x,0} \qquad F_y = \frac{\mu_y}{\mu_{y,act}} F_{y,0}$$

For aligning moment M_x, rolling resistance M_y and aligning moment M_z the formulae (76) until and including (85) are used with $S_{Vyk} = 0$.

Transient Behavior in PAC2002

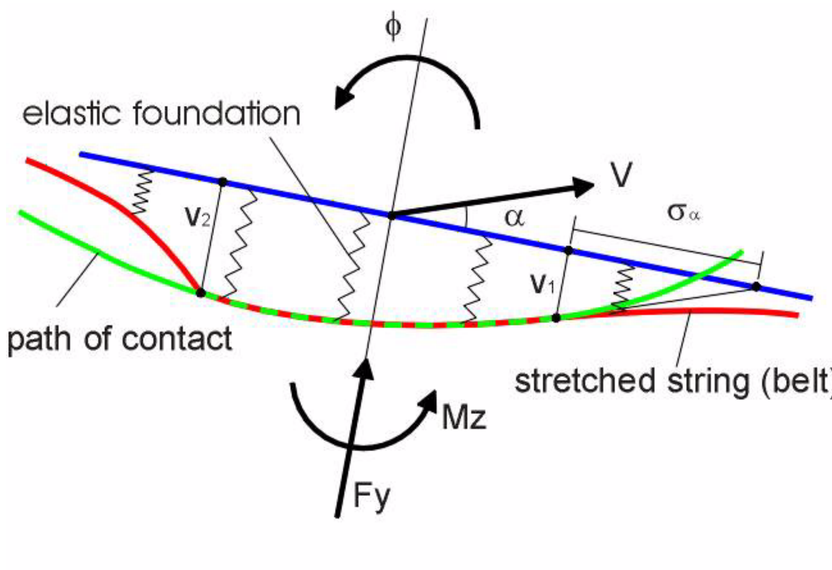
The previous Magic Formula equations are valid for steady-state tire behavior. When driving, however, the tire requires some response time on changes of the inputs. In tire modeling terminology, the low-frequency behavior (up to 15 Hz) is called transient behavior. PAC2002 provides two methods to model transient tire behavior:

- Stretched String
- Contact Mass

Stretched String Model

For accurate transient tire behavior, you can use the stretched string tire model (see reference [1]). The tire belt is modeled as stretched string, which is supported to the rim with lateral (and longitudinal) springs. Stretched String Model for Transient Tire Behavior shows a top-view of the string model. When rolling, the first point having contact with the road adheres to the road (no sliding assumed). Therefore, a lateral deflection of the string arises that depends on the slip angle size and the history of the lateral deflection of previous points having contact with the road.

Stretched String Model for Transient Tire Behavior



For calculating the lateral deflection v_1 of the string in the first point of contact with the road, the following differential equation is valid:

$$\frac{1}{V_x}\frac{dv_1}{dt} + \frac{v_1}{\sigma_\alpha} = tan(\alpha) + a\phi \tag{89}$$

with the relaxation length σ_α in the lateral direction. The turnslip ϕ can be neglected at radii larger than 10 m. This differential equation cannot be used at zero speed, but when multiplying with V_x, the equation can be transformed to:

$$\sigma_\alpha\frac{dv_1}{dt} + |V_x|v_1 = \sigma_\alpha V_{sy} \tag{90}$$

When the tire is rolling, the lateral deflection depends on the lateral slip speed; at standstill, the deflection depends on the relaxation length, which is a measure for the lateral stiffness of the tire. Therefore, with this approach, the tire is responding to a slip speed when rolling and behaving like a spring at standstill.

A similar approach yields the following for the deflection of the string in longitudinal direction:

$$\sigma_\kappa\frac{du_1}{dt} + |V_x|u_1 = \sigma_\kappa V_{sx} \tag{91}$$

Both the longitudinal and lateral relaxation length are defined as of the vertical load:

$$\sigma_\kappa = F_z \cdot (p_{Tx1} + p_{Tx2} df_z) \cdot exp(p_{Tx3} df_z) \cdot (R_0 / F_{z0}) \lambda_{\sigma\kappa} \quad (92)$$

$$\sigma_\alpha = p_{Ty1} sin\left[2arctan\left\{\frac{F_z}{(p_{Ty2} F_{z0} \lambda_{F_{z0}})}\right\}\right] \cdot (1 - p_{Ky3}|\gamma_y|) \cdot R_0 \lambda_{F_{z0}} \cdot \lambda_{\sigma\alpha} \quad (93)$$

Now the practical slip quantities, κ and α, are defined based on the tire deformation:

$$\kappa' = \frac{u_1}{\sigma_x} \cdot sin(V_x) \quad (94)$$

$$\alpha' = atan\left(\frac{v_1}{\sigma_\alpha}\right) \quad (95)$$

Using these practical slip quantities, κ and α, the Magic Formula equations can be used to calculate the tire-road interaction forces and moments:

$$F_x = F_x(\alpha', \kappa', F_z) \quad (96)$$

$$F_y = F_y(\alpha', \kappa', \gamma, F_z) \quad (97)$$

$$M'_z = M'_z(\alpha', \kappa', \gamma, F_z) \quad (98)$$

$$M_z' = M_z'(\alpha', \kappa', F_z) \quad (99)$$

Coefficients and Transient Response

Name:	Name used in tire property file:	Explanation:
p_{Tx1}	PTX1	Longitudinal relaxation length at Fznom
p_{Tx2}	PTX2	Variation of longitudinal relaxation length with load
p_{Tx3}	PTX3	Variation of longitudinal relaxation length with exponent of load
p_{Ty1}	PTY1	Peak value of relaxation length for lateral direction
p_{Ty2}	PTY2	Shape factor for lateral relaxation length
q_{Tz1}	QTZ1	Gyroscopic moment constant
M_{belt}	MBELT	Belt mass of the wheel

Contact Mass Model

The contact mass model is based on the separation of the contact patch slip properties and the tire carcass compliance (see reference [1]). Instead of using relaxation lengths to describe compliance effects, the carcass springs are explicitly incorporated in the model. The contact patch is given some inertia to ensure computational causality. This modeling approach automatically accounts for the lagged response to slip and load changes that diminish at higher levels of slip. The contact patch itself uses relaxation lengths to handle simulations at low speed.

The contact patch can deflect in longitudinal, lateral, and yaw directions with respect to the lower part of the wheel rim. A mass is attached to the contact patch to enable straightforward computations.

The differential equations that govern the dynamics of the contact patch body are:

$$m_c(\dot{V}_{cx} - V_{cy}\dot{\psi}_c) + k_x\dot{u} + c_x u = F_x$$

$$m_c(\dot{V}_{cy} - V_{cx}\dot{\psi}_c) + k_y\dot{u} + c_y u = F_y$$

$$J_c\ddot{\psi}_c + k_\psi\dot{\beta} + c_\psi\beta = M_z$$

The contact patch body with mass m_c and inertia J_c is connected to the wheel through springs c_x, c_y, *and* $c\psi$ and dampers k_x, k_y, and $k\psi$ in longitudinal, lateral, and yaw direction, respectively.

The additional equations for the longitudinal u, lateral v, and yaw β deflections are:

$$\dot{u} = V_{cx} - V_{sx}$$

$$\dot{v} = V_{cy} - V_{sy}$$

$$\dot{\beta} = \dot{\psi}_c - \psi$$

in which V_{cx}, V_{cy} and $\dot{\psi}_c$ are the sliding velocity of the contact body in longitudinal, lateral, and yaw directions, respectively. V_{sx}, V_{sy}, and $\dot{\psi}$ are the corresponding velocities of the lower part of the wheel.

The transient slip equations for side slip, turn-slip, and camber are:

$$\sigma_c\frac{d}{dt}\alpha' + |V_x|\alpha' = V_{cy} - V_x\beta + |V_x|\beta_{st}$$

$$\sigma_c\frac{d\alpha'_t}{dt} + |V_x|\alpha'_t = |V_x|\alpha'$$

$$\sigma_c \frac{d\varphi'_c}{dt} + |V_x|\varphi'_c = \dot{\psi}_\gamma$$

$$\sigma_{F2} \frac{d\varphi'_{F2}}{dt} + |V_x|\varphi'_{cF2} = \dot{\psi}_\gamma$$

$$\sigma_{\varphi 1} \frac{d\varphi'_1}{dt} + |V_x|\varphi'_1 = \dot{\psi}_\gamma$$

$$\sigma_{\varphi 2} \frac{d\varphi'_2}{dt} + |V_x|\varphi'_2 = \dot{\psi}_\gamma$$

where the calculated deflection angle has been used:

$$\beta_{st} = \frac{M_z}{c_\phi}$$

The tire total spin velocity is:

$$\dot{\psi}_\gamma = \psi_c - (1 - \varepsilon_\gamma)\Omega sin\gamma$$

With the transient slip equations, the composite transient turn-slip quantities are calculated:

$$\varphi'_F = 2\varphi'_c - \varphi'_{F2}$$

$$\varphi'_M = \varepsilon_\varphi \varphi'_c + \varepsilon_{\varphi 12}(\varphi'_1 - \varphi'_2)$$

The tire forces are calculated with φ'_F and the tire moments with φ'_M.

The relaxation lengths are reduced with slip:

$$\sigma_c = a \cdot (1 - \theta_\zeta)$$

$$\sigma_2 = \frac{t_0}{a}\sigma_c$$

$$\sigma_{F2} = b_{F2}\sigma_c$$

$$\sigma_{\varphi 1} = b_{\varphi 1}\sigma_c$$

$$\sigma_{\varphi 2} = b_{\varphi 2}\sigma_c$$

Here a is half the contact length according to:

$$a = p_{A1} R_0 \left(\frac{\rho_z}{R_0} + p_{A2} \sqrt{\frac{\rho_z}{R_0}} \right)$$

The composite tire parameter reads:

$$\theta = \frac{K_{y0}}{3 \mu_y F_x}$$

and the equivalent slip:

$$\zeta = \frac{1}{1 + \kappa'} \sqrt{\{|\alpha'| + a \varepsilon_{\varphi 12} |\varphi'_1 - \varphi'_2|\}^2 + \left(\frac{K_{x0}}{K_{y0}}\right)^2 \left\{|\kappa'| + \frac{2}{3} b |\varphi'_c|\right\}^2}$$

Coefficients and Transient Response

Name:	Name used in tire property file:	Explanation:
m_c	MC	Contact body mass
I_c	IC	Contact body moment of inertia
k_x	KX	Longitudinal damping
k_y	KY	Lateral damping
$k\varphi$	KP	Yaw damping
c_x	CX	Longitudinal stiffness
c_y	CY	Lateral stiffness
$c\varphi$	CP	Yaw stiffness
p_{A1}	PA1	Half contact length with vertical tire deflection
p_{A2}	PA2	Half contact length with square root of vertical tire deflection
ε_φ	EP	Composite turn-slip (moment)
$\varepsilon_{\varphi 12}$	EP12	Composite turn-slip (moment) increment
b_{F2}	BF2	Second relaxation length factor
$b\varphi 1$	BP1	First moment relaxation length factor
$b\varphi 2$	BP2	Second moment relaxation length factor

The remaining contact mass model parameters are estimated automatically based on longitudinal and lateral stiffness specified in the tire property file.

Gyroscopic Couple in PAC2002

When having fast rotations about the vertical axis in the wheel plane, the inertia of the tire belt may lead to gyroscopic effects. To cope with this additional moment, the following contribution is added to the total aligning moment:

$$M_{z,\,gyr} = c_{gyr} m_{belt} V_{rl} \frac{dv}{dt} \cos[\arctan(B_r \alpha_{r,\,eq})] \tag{100}$$

with the parameter (in addition to the basic tire parameter m_{belt}):

$$c_{gyr} = q_{Tz1} \cdot \lambda_{gyr} \tag{101}$$

and:

$$\cos[\arctan(B_r \alpha_{r,\,eq})] = 1 \tag{102}$$

The total aligning moment now becomes:

$$M_z = M'_z + M_{z,\,gyr} \tag{103}$$

Coefficients and Transient Response

Name:	Name used in tire property file:	Explanation:
p_{Tx1}	PTX1	Longitudinal relaxation length at Fznom
p_{Tx2}	PTX2	Variation of longitudinal relaxation length with load
p_{Tx3}	PTX3	Variation of longitudinal relaxation length with exponent of load
p_{Ty1}	PTY1	Peak value of relaxation length for lateral direction
p_{Ty2}	PTY2	Shape factor for lateral relaxation length
q_{Tz1}	QTZ1	Gyroscopic moment constant
M_{belt}	MBELT	Belt mass of the wheel

Left and Right Side Tires

In general, a tire produces a lateral force and aligning moment at zero slip angle due to the tire construction, known as conicity and plysteer. In addition, the tire characteristics cannot be symmetric for positive and negative slip angles.

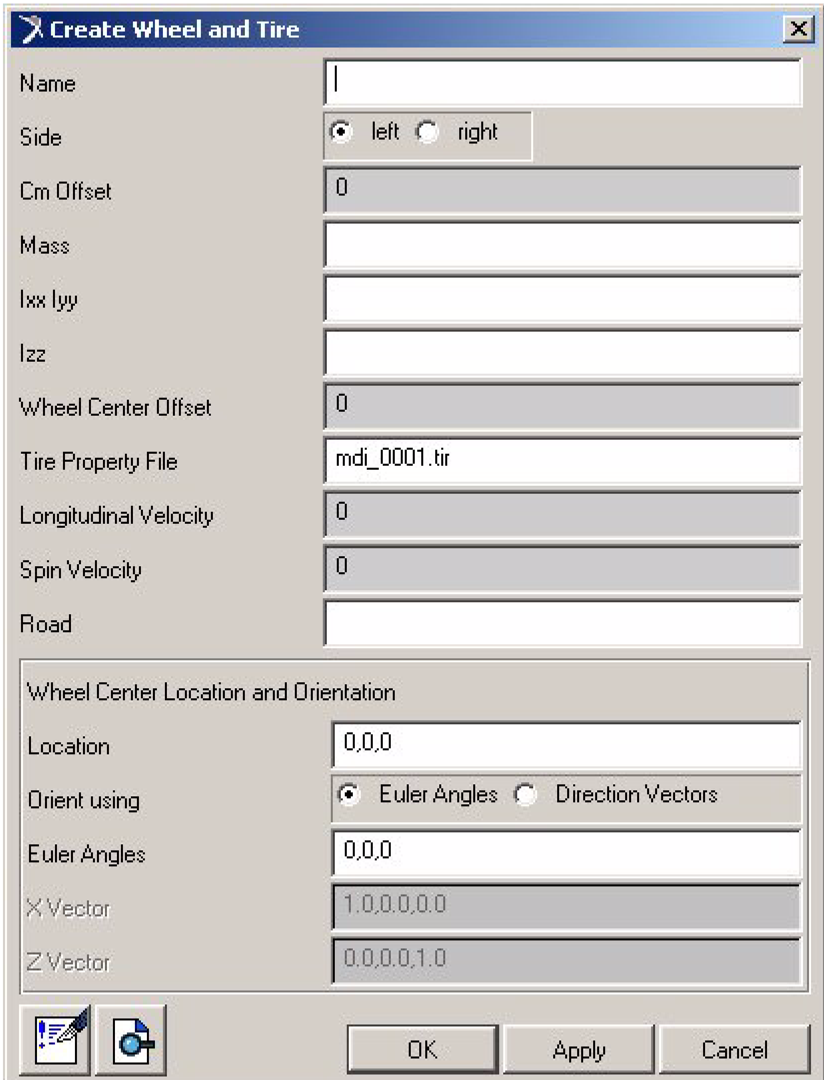
A tire property file with the parameters for the model results from testing with a tire that is mounted in a tire test bench comparable either to the left or the right side of a vehicle. If these coefficients are used for both the left and the right side of the vehicle model, the vehicle does not drive straight at zero steering wheel angle.

The latest versions of tire property files contain a keyword TYRESIDE in the [MODEL] section that indicates for which side of the vehicle the tire parameters in that file are valid (TYRESIDE = 'LEFT' or TYRESIDE = 'RIGHT'). .

If this keyword is available, Adams/Car corrects for the conicity and plysteer and asymmetry when using a tire property file on the opposite side of the vehicle. In fact, the tire characteristics are mirrored with respect to slip angle zero. In Adams/View, this option can only be used when the tire is generated by the graphical user interface: select **Build -> Forces -> Special Force: Tire**.

Next to the LEFT and RIGHT side option of TYRESIDE, you can also set SYMMETRIC: then the tire characteristics are modified during initialization to show symmetric performance for left and right side corners and zero conicity and plysteer (no offsets).Also, when you set the tire property file to SYMMETRIC, the tire characteristics are changed to symmetric behavior.

Create Wheel and Tire Dialog Box in Adams/View



USE_MODES of PAC2002: from Simple to Complex

The parameter USE_MODE in the tire property file allows you to switch the output of the PAC2002 tire model from very simple (that is, steady-state cornering) to complex (transient combined cornering and braking).

The options for the USE_MODE and the output of the model have been listed in the table below.

USE_MODE Values of PAC2002 and Related Tire Model Output

USE_MODE:	State:	Slip conditions:	PAC2002 output (forces and moments):
0	Steady state	Acts as a vertical spring & damper	0, 0, F_z, 0, 0, 0
1	Steady state	Pure longitudinal slip	F_x, 0, F_z, 0, M_y, 0
2	Steady state	Pure lateral (cornering) slip	0, F_y, F_z, M_x, 0, M_z
3	Steady state	Longitudinal and lateral (not combined)	F_x, F_y, F_z, M_x, M_y, M_z
4	Steady state	Combined slip	F_x, F_y, F_z, M_x, M_y, M_z
11	Transient	Pure longitudinal slip	F_x, 0, F_z, 0, M_y, 0
12	Transient	Pure lateral (cornering) slip	0, F_y, F_z, M_x, 0, M_z
13	Transient	Longitudinal and lateral (not combined)	F_x, F_y, F_z, M_x, M_y, M_z
14	Transient	Combined slip	F_x, F_y, F_z, M_x, M_y, M_z
15	Transient	Combined slip and turn-slip	F_x, F_y, F_z, M_x, M_y, M_z
21	Advanced transient	Pure longitudinal slip	F_x, 0, F_z, M_y, 0
22	Advanced transient	Pure lateral (cornering slip)	0, F_y, F_z, M_x, 0, M_z
23	Advanced transient	Longitudinal and lateral (not combined)	F_x, F_y, F_z, M_x, M_y, M_z
24	Advanced transient	Combined slip	F_x, F_y, F_z, M_x, M_y, M_z
25	Advanced transient	Combined slip and turn-slip/parking	F_x, F_y, F_z, M_x, M_y, M_z

Quality Checks for the Tire Model Parameters

Because PAC2002 uses an empirical approach to describe tire - road interaction forces, incorrect parameters can easily result in non-realistic tire behavior. Below is a list of the most important items to ensure the quality of the parameters in a tire property file:

- Rolling Resistance
- Camber (Inclination) Effects
- Validity Range of the Tire Model Input

Note: **Do not change F_{z0} (FNOMIN) and R_0 (UNLOADED_RADIUS)** in your tire property file. It will change the complete tire characteristics because these two parameters are used to make all parameters without dimension.

Rolling Resistance

For a realistic rolling resistance, the parameter q_{sy1} must be positive. For car tires, it can be in the order of 0.006 - 0.01 (0.6% - 1.0%); for heavy commercial truck tires, it can be around 0.006 (0.6%).

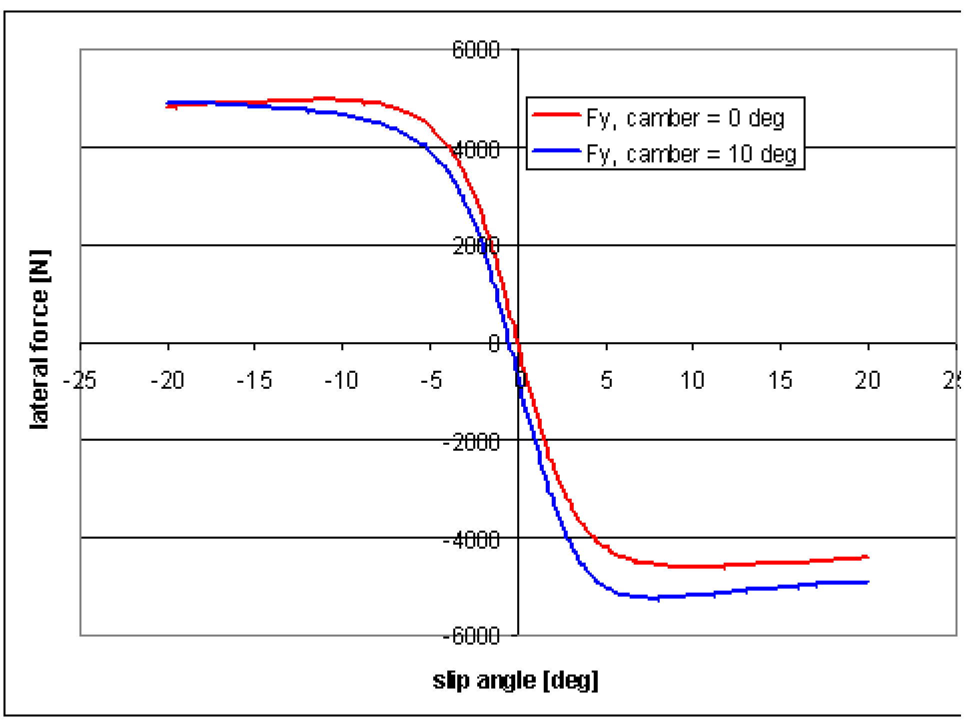
Tire property files with the keyword FITTYP=5 determine the rolling resistance in a different way (see equation (88)). To avoid the 'old' rolling resistance calculation, remove the keyword FITTYP and add a section like the following:

```
$---------------------------------------------------rolling
resistance[ROLLING_COEFFICIENTS]
QSY1 = 0.01
QSY2 = 0
QSY3 = 0
QSY4 = 0
```

Camber (Inclination) Effects

Camber stiffness has not been explicitly defined in PAC2002; however, for car tires, positive inclination should result in a negative lateral force at zero slip angle. If positive inclination results in an increase of the lateral force, the coefficient may not be valid for the ISO but for the SAE coordinate system. Note that PAC2002 only uses coefficients for the TYDEX W-axis (ISO) system.

Effect of Positive Camber on the Lateral Force in TYDEX W-axis (ISO) System



The table below lists further checks on the PAC2002 parameters.

Checklist for PAC2002 Parameters and Properties

Parameter/property:	Requirement:	Explanation:
LONGVL	1 m/s	Reference velocity at which parameters are measured
VXLOW	Approximately 1 m/s	Threshold for scaling down forces and moments
D_x	> 0	Peak friction (see equation (22))
p_{Dx1}/p_{Dx2}	< 0	Peak friction F_x must decrease with increasing load
K_x	> 0	Long slip stiffness (see equation (25))
D_y	> 0	Peak friction (see equation (34))
p_{Dy1}/p_{Dy2}	< 0	Peak friction F_x must decrease with increasing load
K_y	< 0	Cornering stiffness (see equation (37))
q_{sy1}	> 0	Rolling resistance, in the range of 0.005 - 0.015

Validity Range of the Tire Model Input

In the tire property file, a range of the input variables has been given in which the tire properties are supposed to be valid. These validity range parameters are (the listed values can be different):

```
$---------------------------------------------------long_slip_range
[LONG_SLIP_RANGE]
KPUMIN     = -1.5       $Minimum valid wheel slip
KPUMAX     = 1.5        $Maximum valid wheel slip
$--------------------------------------------------slip_angle_range
[SLIP_ANGLE_RANGE]
ALPMIN     = -1.5708    $Minimum valid slip angle
ALPMAX     = 1.5708     $Maximum valid slip angle
$---------------------------------------------inclination_slip_range
[INCLINATION_ANGLE_RANGE]
CAMMIN     = -0.26181   $Minimum valid camber angle
CAMMAX     = 0.26181    $Maximum valid camber angle
$-----------------------------------------------vertical_force_range
[VERTICAL_FORCE_RANGE]
FZMIN      = 225        $Minimum allowed wheel load
FZMAX      = 10125      $Maximum allowed wheel load
```

If one of the input parameters exceeds a minimum or maximum validity value, the calculation in the tire model is performed with the minimum or maximum value of this range to avoid non-realistic tire behavior. In that case, a message appears warning you that one of the inputs exceeds a validity value.

Standard Tire Interface (STI) for PAC2002

Because all Adams products use the Standard Tire Interface (STI) for linking the tire models to Adams/Solver, below is a brief background of the STI history (see also reference [4]).

At the First International Colloquium on Tire Models for Vehicle Dynamics Analysis on October 21-22, 1991, the International Tire Workshop working group was established (TYDEX).

The working group concentrated on tire measurements and tire models used for vehicle simulation purposes. For most vehicle dynamics studies, people used to develop their own tire models. Because all car manufacturers and their tire suppliers have the same goal (that is, development of tires to improve dynamic safety of the vehicle) it aimed for standardization in tire behavior description.

In TYDEX, two expert groups, consisting of participants of vehicle industry (passenger cars and trucks), tire manufacturers, other suppliers and research laboratories, had been defined with following goals:

- The first expert group's (Tire Measurements - Tire Modeling) main goal was to specify an interface between tire measurements and tire models. The result was the TYDEX-Format [2] to describe tire measurement data.
- The second expert group's (Tire Modeling - Vehicle Modeling) main goal was to specify an interface between tire models and simulation tools, which resulted in the Standard Tire Interface (STI) [3]. The use of this interface should ensure that a wide range of simulation software can be linked to a wide range of tire modeling software.

Definitions

- General
- Tire Kinematics
- Slip Quantities
- Force and Moments

General

General Definitions

Term:	Definition:
Road tangent plane	Plane with the normal unit vector (tangent to the road) in the tire-road contact point C.
C-axis system	Coordinate system mounted on the wheel carrier at the wheel center according to TYDEX, ISO orientation.
Wheel plane	The plane in the wheel center that is formed by the wheel when considered a rigid disc with zero width.

Term:	Definition:
Contact point C	Contact point between tire and road, defined as the intersection of the wheel plane and the projection of the wheel axis onto the road plane.
W-axis system	Coordinate system at the tire contact point C, according to TYDEX, ISO orientation.

Tire Kinematics

Tire Kinematics Definitions

Parameter:	Definition:	Units:
R_0	Unloaded tire radius	[m]
R	Loaded tire radius	[m]
R_e	Effective tire radius	[m]
ρ	Radial tire deflection	[m]
ρ^{d}	Dimensionless radial tire deflection	[-]
ρ_{FZ0}	Radial tire deflection at nominal load	[m]
m_{belt}	Tire belt mass	[kg]
Ω	Rotational velocity of the wheel	$[rads^{-1}]$

Slip Quantities

Slip Quantities Definitions

Parameter:	Definition:	Units:
V	Vehicle speed	$[ms^{-1}]$
V_{sx}	Slip speed in x direction	$[ms^{-1}]$
V_{sy}	Slip speed in y direction	$[ms^{-1}]$
V_s	Resulting slip speed	$[ms^{-1}]$
V_x	Rolling speed in x direction	$[ms^{-1}]$
V_y	Lateral speed of tire contact center	$[ms^{-1}]$
V_r	Linear speed of rolling	$[ms^{-1}]$
κ	Longitudinal slip	[-]
α	Slip angle	[rad]
γ	Inclination angle	[rad]

Forces and Moments

Force and Moment Definitions

Abbreviation:	Definition:	Units:
F_z	Vertical wheel load	[N]
F_{z0}	Nominal load	[N]
df_z	Dimensionless vertical load	[-]
F_x	Longitudinal force	[N]
F_y	Lateral force	[N]
M_x	Overturning moment	[Nm]
M_y	Braking/driving moment	[Nm]
M_z	Aligning moment	[Nm]

References

1. H.B. Pacejka, Tyre and Vehicle Dynamics, 2002, Butterworth-Heinemann, ISBN 0 7506 5141 5.
2. H.-J. Unrau, J. Zamow, TYDEX-Format, Description and Reference Manual, Release 1.1, Initiated by the International Tire Working Group, July 1995.
3. A. Riedel, Standard Tire Interface, Release 1.2, Initiated by the Tire Workgroup, June 1995.
4. J.J.M. van Oosten, H.-J. Unrau, G. Riedel, E. Bakker, TYDEX Workshop: Standardisation of Data Exchange in Tyre Testing and Tyre Modelling, Proceedings of the 2nd International Colloquium on Tyre Models for Vehicle Dynamics Analysis, Vehicle System Dynamics, Volume 27, Swets & Zeitlinger, Amsterdam/Lisse, 1996.

Example of PAC2002 Tire Property File

```
[MDI_HEADER]
FILE_TYPE                ='tir'
FILE_VERSION             =3.0
FILE_FORMAT              ='ASCII'
! : TIRE_VERSION :      PAC2002
! : COMMENT :           Tire                     235/60R16
! : COMMENT :           Manufacturer
! : COMMENT :           Nom. section with  (m)   0.235
! : COMMENT :           Nom. aspect ratio  (-)   60
! : COMMENT :           Infl. pressure    (Pa)   200000
! : COMMENT :           Rim radius         (m)   0.19
! : COMMENT :           Measurement ID
! : COMMENT :           Test speed       (m/s)   16.6
! : COMMENT :           Road surface
! : COMMENT :           Road condition           Dry
! : FILE_FORMAT :       ASCII
! : Copyright MSC.Software, Fri Jan 23 14:30:06 2004
!
! USE_MODE specifies the type of calculation performed:
!       0: Fz only, no Magic Formula evaluation
```

```
!       1: Fx,My only
!       2: Fy,Mx,Mz only
!       3: Fx,Fy,Mx,My,Mz uncombined force/moment calculation
!       4: Fx,Fy,Mx,My,Mz combined force/moment calculation
!     +10: including relaxation behaviour
!     *-1: mirroring of tyre characteristics
!
!    example: USE_MODE = -12 implies:
!       -calculation of Fy,Mx,Mz only
!       -including relaxation effects
!       -mirrored tyre characteristics
!
$----------------------------------------------------------------units
[UNITS]
LENGTH                  ='meter'
FORCE                   ='newton'
ANGLE                   ='radians'
MASS                    ='kg'
TIME                    ='second'
$----------------------------------------------------------------model
[MODEL]
PROPERTY_FILE_FORMAT    ='PAC2002'
USE_MODE                = 14                  $Tyre use switch (IUSED)
VXLOW                   = 1
LONGVL                  = 16.6                $Measurement speed
TYRESIDE                = 'LEFT'              $Mounted side of tyre at
vehicle/test bench
$-----------------------------------------------------------dimensions
[DIMENSION]
UNLOADED_RADIUS         = 0.344               $Free tyre radius
WIDTH                  = 0.235               $Nominal section width of the
tyre
ASPECT_RATIO            = 0.6                 $Nominal aspect ratio
RIM_RADIUS              = 0.19                $Nominal rim radius
RIM_WIDTH               = 0.16                $Rim width
$------------------------------------------------------------parameter
[VERTICAL]
VERTICAL_STIFFNESS     = 2.1e+005           $Tyre vertical stiffness
VERTICAL_DAMPING       = 50                  $Tyre vertical damping
BREFF                = 8.4                 $Low load stiffness e.r.r.
DREFF                  = 0.27                $Peak value of e.r.r.
FREFF                = 0.07                $High load stiffness e.r.r.
FNOMIN                  = 4850                $Nominal wheel load
$-------------------------------------------------------long_slip_range
[LONG_SLIP_RANGE]
KPUMIN                = -1.5                 $Minimum valid wheel slip
KPUMAX                = 1.5                  $Maximum valid wheel slip
$------------------------------------------------------slip_angle_range
[SLIP_ANGLE_RANGE]
ALPMIN                = -1.5708              $Minimum valid slip angle
ALPMAX                = 1.5708               $Maximum valid slip angle
$-----------------------------------------------inclination_slip_range
[INCLINATION_ANGLE_RANGE]
CAMMIN                = -0.26181            $Minimum valid camber angle
CAMMAX                = 0.26181             $Maximum valid camber angle
$-------------------------------------------------vertical_force_range
[VERTICAL_FORCE_RANGE]
FZMIN                 = 225                 $Minimum allowed wheel load
FZMAX                 = 10125               $Maximum allowed wheel load
$---------------------------------------------------------------scaling
```

```
[SCALING_COEFFICIENTS]
LFZO     = 1    $Scale factor of nominal (rated) load
LCX      = 1    $Scale factor of Fx shape factor
LMUX     = 1    $Scale factor of Fx peak friction coefficient
LEX      = 1    $Scale factor of Fx curvature factor
LKX      = 1    $Scale factor of Fx slip stiffness
LHX      = 1    $Scale factor of Fx horizontal shift
LVX      = 1    $Scale factor of Fx vertical shift
LGAX     = 1    $Scale factor of camber for Fx
LCY      = 1    $Scale factor of Fy shape factor
LMUY     = 1    $Scale factor of Fy peak friction coefficient
LEY      = 1    $Scale factor of Fy curvature factor
LKY      = 1    $Scale factor of Fy cornering stiffness
LHY      = 1    $Scale factor of Fy horizontal shift
LVY      = 1    $Scale factor of Fy vertical shift
LGAY     = 1    $Scale factor of camber for Fy
LTR      = 1    $Scale factor of Peak of pneumatic trail
LRES     = 1    $Scale factor for offset of residual torque
LGAZ     = 1    $Scale factor of camber for Mz
LXAL     = 1    $Scale factor of alpha influence on Fx
LYKA     = 1    $Scale factor of alpha influence on Fx
LVYKA    = 1    $Scale factor of kappa induced Fy
LS       = 1    $Scale factor of Moment arm of Fx
LSGKP    = 1    $Scale factor of Relaxation length of Fx
LSGAL    = 1    $Scale factor of Relaxation length of Fy
LGYR     = 1    $Scale factor of gyroscopic torque
LMX      = 1    $Scale factor of overturning couple
LVMX     = 1    $Scale factor of Mx vertical shift
LMY      = 1    $Scale factor of rolling resistance torque
$-------------------------------------------------------longitudinal
[LONGITUDINAL_COEFFICIENTS]
PCX1      = 1.6411         $Shape factor Cfx for longitudinal force
PDX1      = 1.1739         $Longitudinal friction Mux at Fznom
PDX2      = -0.16395       $Variation of friction Mux with load
PDX3      = 0              $Variation of friction Mux with camber
PEX1      = 0.46403        $Longitudinal curvature Efx at Fznom
PEX2      = 0.25022        $Variation of curvature Efx with load
PEX3      = 0.067842       $Variation of curvature Efx with load squared
PEX4      = -3.7604e-005   $Factor in curvature Efx while driving
PKX1      = 22.303         $Longitudinal slip stiffness Kfx/Fz at Fznom
PKX2      = 0.48896        $Variation of slip stiffness Kfx/Fz with load
PKX3      = 0.21253        $Exponent in slip stiffness Kfx/Fz with load
PHX1      = 0.0012297      $Horizontal shift Shx at Fznom
PHX2      = 0.0004318      $Variation of shift Shx with load
PVX1      = -8.8098e-006   $Vertical shift Svx/Fz at Fznom
PVX2      = 1.862e-005     $Variation of shift Svx/Fz with load
RBX1      = 13.276         $Slope factor for combined slip Fx reduction
RBX2      = -13.778        $Variation of slope Fx reduction with kappa
RCX1      = 1.2568         $Shape factor for combined slip Fx reduction
REX1      = 0.65225        $Curvature factor of combined Fx
REX2      = -0.24948       $Curvature factor of combined Fx with load
RHX1      = 0.0050722      $Shift factor for combined slip Fx reduction
PTX1      = 2.3657         $Relaxation length SigKap0/Fz at Fznom
PTX2      = 1.4112         $Variation of SigKap0/Fz with load
PTX3      = 0.56626        $Variation of SigKap0/Fz with exponent of load
$--------------------------------------------------------overturning
[OVERTURNING_COEFFICIENTS]
QSX1     = 0     $Lateral force induced overturning moment
QSX2     = 0     $Camber induced overturning couple
QSX3     = 0     $Fy induced overturning couple
```

```
$---------------------------------------------------------------lateral
[LATERAL_COEFFICIENTS]
PCY1    = 1.3507         $Shape factor Cfy for lateral forces
PDY1    = 1.0489         $Lateral friction Muy
PDY2    = -0.18033       $Variation of friction Muy with load
PDY3    = -2.8821        $Variation of friction Muy with squared camber
PEY1    = -0.0074722     $Lateral curvature Efy at Fznom
PEY2    = -0.0063208     $Variation of curvature Efy with load
PEY3    = -9.9935        $Zero order camber dependency of curvature Efy
PEY4    = -760.14        $Variation of curvature Efy with camber
PKY1    = -21.92         $Maximum value of stiffness Kfy/Fznom
PKY2    = 2.0012         $Load at which Kfy reaches maximum value
PKY3    = -0.024778      $Variation of Kfy/Fznom with camber
PHY1    = 0.0026747      $Horizontal shift Shy at Fznom
PHY2    = 8.9094e-005    $Variation of shift Shy with load
PHY3    = 0.031415       $Variation of shift Shy with camber
PVY1    = 0.037318       $Vertical shift in Svy/Fz at Fznom
PVY2    = -0.010049      $Variation of shift Svy/Fz with load
PVY3    = -0.32931       $Variation of shift Svy/Fz with camber
PVY4    = -0.69553       $Variation of shift Svy/Fz with camber and load
RBY1    = 7.1433         $Slope factor for combined Fy reduction
RBY2    = 9.1916         $Variation of slope Fy reduction with alpha
RBY3    = -0.027856      $Shift term for alpha in slope Fy reduction
RCY1    = 1.0719         $Shape factor for combined Fy reduction
REY1    = -0.27572       $Curvature factor of combined Fy
REY2    = 0.32802        $Curvature factor of combined Fy with load
RHY1    = 5.7448e-006    $Shift factor for combined Fy reduction
RHY2    = -3.1368e-005   $Shift factor for combined Fy reduction with load
RVY1    = -0.027825      $Kappa induced side force Svyk/Muy*Fz at Fznom
RVY2    = 0.053604       $Variation of Svyk/Muy*Fz with load
RVY3    = -0.27568       $Variation of Svyk/Muy*Fz with camber
RVY4    = 12.12          $Variation of Svyk/Muy*Fz with alpha
RVY5    = 1.9            $Variation of Svyk/Muy*Fz with kappa
RVY6    = -10.704        $Variation of Svyk/Muy*Fz with atan(kappa)
PTY1    = 2.1439         $Peak value of relaxation length SigAlp0/R0
PTY2    = 1.9829         $Value of Fz/Fznom where SigAlp0 is extreme
$--------------------------------------------------rolling resistance
[ROLLING_COEFFICIENTS]
QSY1     = 0.01           $Rolling resistance torque coefficient
QSY2     = 0              $Rolling resistance torque depending on Fx
QSY3     = 0              $Rolling resistance torque depending on speed
QSY4     = 0              $Rolling resistance torque depending on speed ^4
$------------------------------------------------------------aligning
[ALIGNING_COEFFICIENTS]
QBZ1    = 10.904         $Trail slope factor for trail Bpt at Fznom
QBZ2    = -1.8412        $Variation of slope Bpt with load
QBZ3    = -0.52041       $Variation of slope Bpt with load squared
QBZ4    = 0.039211       $Variation of slope Bpt with camber
QBZ5    = 0.41511        $Variation of slope Bpt with absolute camber
QBZ9    = 8.9846         $Slope factor Br of residual torque Mzr
QBZ10   = 0              $Slope factor Br of residual torque Mzr
QCZ1    = 1.2136         $Shape factor Cpt for pneumatic trail
QDZ1    = 0.093509       $Peak trail Dpt" = Dpt*(Fz/Fznom*R0)
QDZ2    = -0.0092183     $Variation of peak Dpt" with load
QDZ3    = -0.057061      $Variation of peak Dpt" with camber
QDZ4    = 0.73954        $Variation of peak Dpt" with camber squared
QDZ6    = -0.0067783     $Peak residual torque Dmr" = Dmr/(Fz*R0)
QDZ7    = 0.0052254      $Variation of peak factor Dmr" with load
QDZ8    = -0.18175       $Variation of peak factor Dmr" with camber
QDZ9    = 0.029952       $Variation of peak factor Dmr" with camber and load
```

```
QEZ1    = -1.5697       $Trail curvature Ept at Fznom
QEZ2    = 0.33394       $Variation of curvature Ept with load
QEZ3    = 0             $Variation of curvature Ept with load squared
QEZ4    = 0.26711       $Variation of curvature Ept with sign of Alpha-t
QEZ5    = -3.594        $Variation of Ept with camber and sign Alpha-t
QHZ1    = 0.0047326     $Trail horizontal shift Sht at Fznom
QHZ2    = 0.0026687     $Variation of shift Sht with load
QHZ3    = 0.11998       $Variation of shift Sht with camber
QHZ4    = 0.059083      $Variation of shift Sht with camber and load
SSZ1    = 0.033372      $Nominal value of s/R0: effect of Fx on Mz
SSZ2    = 0.0043624     $Variation of distance s/R0 with Fy/Fznom
SSZ3    = 0.56742       $Variation of distance s/R0 with camber
SSZ4    = -0.24116      $Variation of distance s/R0 with load and camber
QTZ1    = 0.2           $Gyration torque constant
MBELT   = 5.4           $Belt mass of the wheel
$------------------------------------------------turn-slip parameters
[TURNSLIP_COEFFICIENTS]
PECP1     = 0.7        $Camber stiffness reduction factor
PECP2     = 0.0        $Camber stiffness reduction factor with load
PDXP1     = 0.4        $Peak Fx reduction due to spin
PDXP2     = 0.0        $Peak Fx reduction due to spin with load
PDXP3     = 0.0        $Peak Fx reduction due to spin with longitudinal slip
PDYP1     = 0.4        $Peak Fy reduction due to spin
PDYP2     = 0.0        $Peak Fy reduction due to spin with load
PDYP3     = 0.0        $Peak Fy reduction due to spin with lateral slip
PDYP4     = 0.0        $Peak Fy reduction with square root of spin
PKYP1     = 1.0        $Cornering stiffness reduction due to spin
PHYP1     = 1.0        $Fy lateral shift shape factor
PHYP2     = 0.15       $Maximum Fy lateral shift
PHYP3     = 0.0        $Maximum Fy lateral shift with load
PHYP4     = -4.0       $Fy lateral shift curvature factor
QDTP1     = 10.0       $Pneumatic trail reduction factor
QBRP1     = 0.1        $Residual torque reduction factor with lateral slip
QCRP1     = 0.2        $Turning moment at constant turning with zero speed
QCRP2     = 0.1        $Turning moment at 90 deg lateral slip
QDRP1     = 1.0        $Maximum turning moment
QDRP2     = -1.5       $Location of maximum turning moment
$---------------------------------------------contact patch parameters
[CONTACT_COEFFICIENTS]
PA1       = 0.4147     $Half contact length dependency on Fz)
PA2       = 1.9129     $Half contact length dependency on sqrt(Fz/R0)
$---------------------------------------------contact patch slip model
[DYNAMIC_COEFFICIENTS]
MC        = 1.0        $Contact mass
IC        = 0.05       $Contact moment of inertia
KX        = 409.0      $Contact longitudinal damping
KY        = 320.8      $Contact lateral damping
KP        = 11.9       $Contact yaw damping
CX        = 4.350e+005 $Contact longitudinal stiffness
CY        = 1.665e+005 $Contact lateral stiffness
CP        = 20319      $Contact yaw stiffness
EP        = 1.0
EP12      = 4.0
BF2       = 0.5
BP1       = 0.5
BP2       = 0.67
$-------------------------------------------------------loaded radius
[LOADED_RADIUS_COEFFICIENTS]
QV1       = 0.000071   $Tire radius growth coefficient
QV2       = 2.489      $Tire stiffness variation coefficient with speed
```

```
QFCX1    = 0.1        $Tire stiffness interaction with Fx
QFCY1    = 0.3        $Tire stiffness interaction with Fy
QFCG1    = 0.0        $Tire stiffness interaction with camber
QFZ1     = 0.0        $Linear stiffness coefficient, if zero,
VERTICAL_STIFFNESS is taken
QFZ2     = 14.35      $Tire vertical stiffness coefficient (quadratic)
```

Contact Methods

The PAC2002 model supports the following roads:

- 2D Roads, see Using the 2D Road Model
- 3D Spline Roads, see Adams/3D Spline Road Model

  Note that the PAC2002 model has only one point of contact with the road; therefore, the wavelength of road obstacles must be longer than the tire radius for realistic output of the model. In addition, the contact force computed by this tire model is normal to the road plane. Therefore, the contact point does not generate a longitudinal force when rolling over a short obstacle, such as a cleat or pothole.
- 3D Shell Roads, see Adams/Tire 3D Shell Road Model

  For ride and comfort analyses, we recommend more sophisticated tire models, such as Ftire.